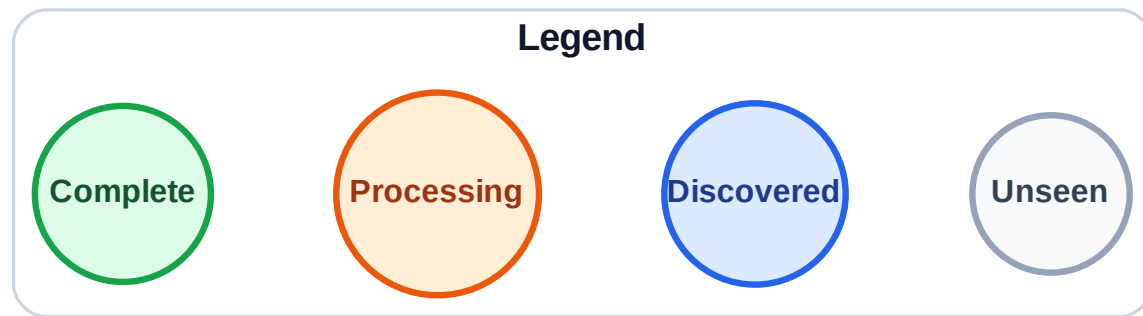
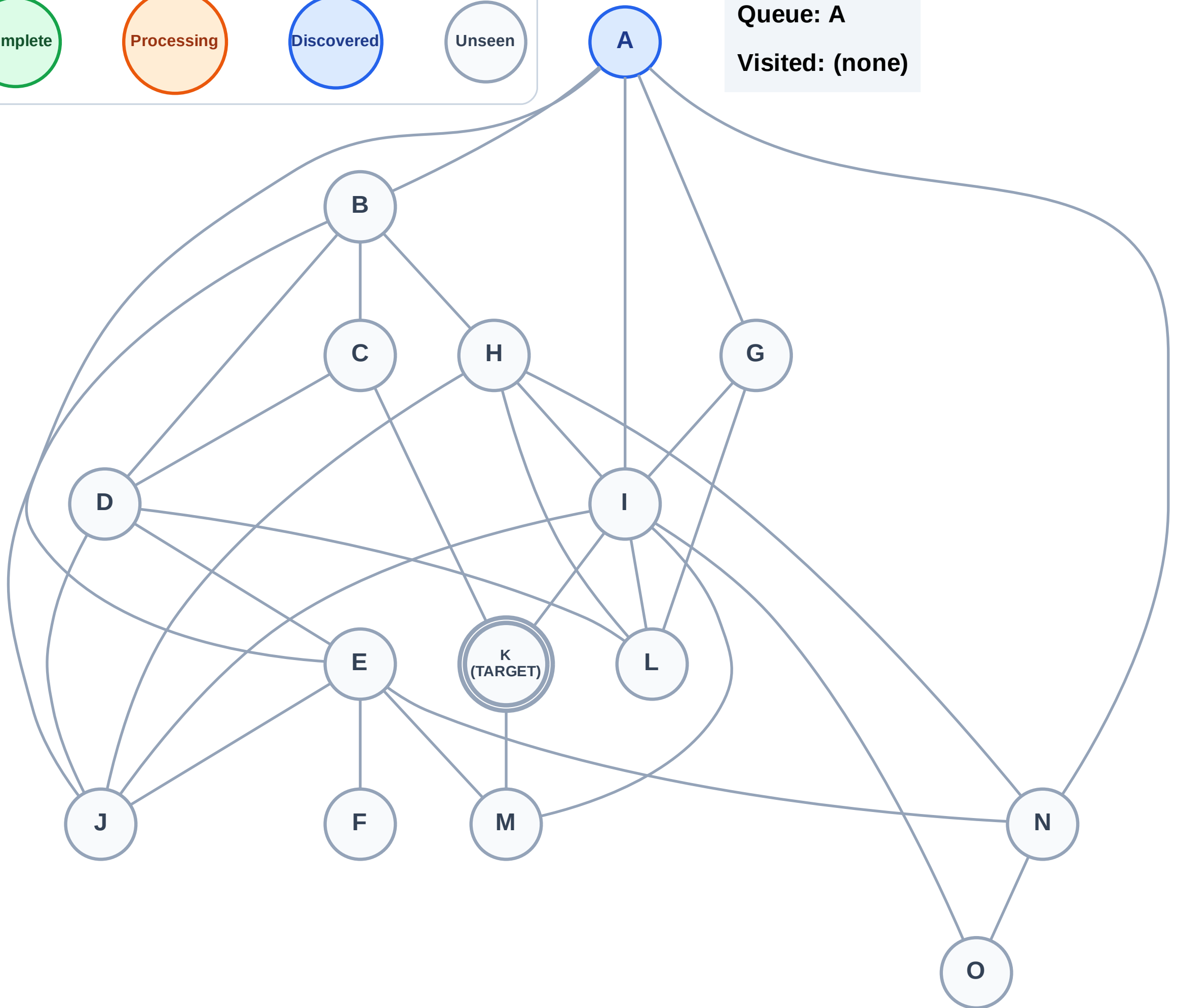


BFS Traversal

Step 1: Start at A (Seeking target node K)



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



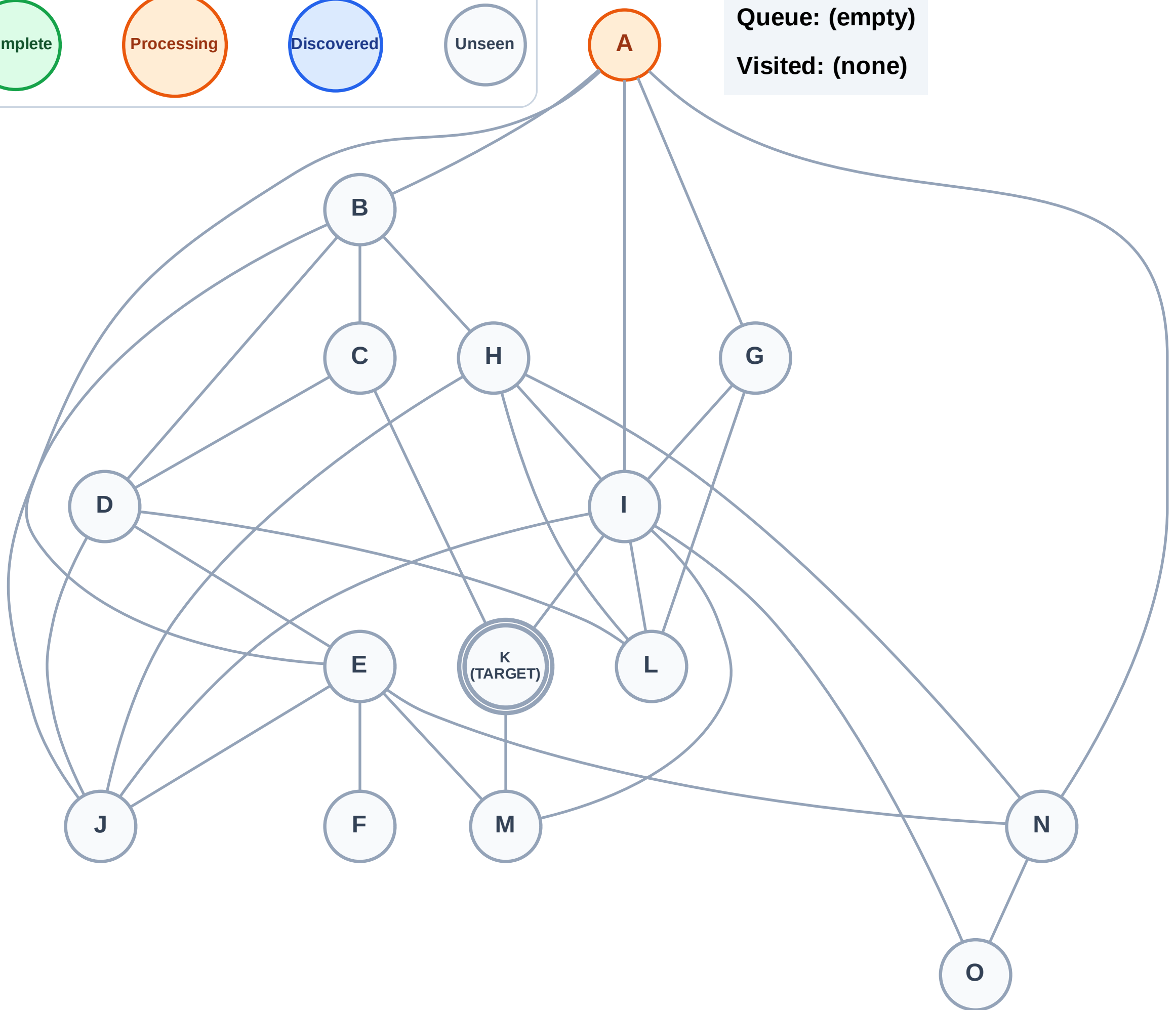
Discovered



Unseen

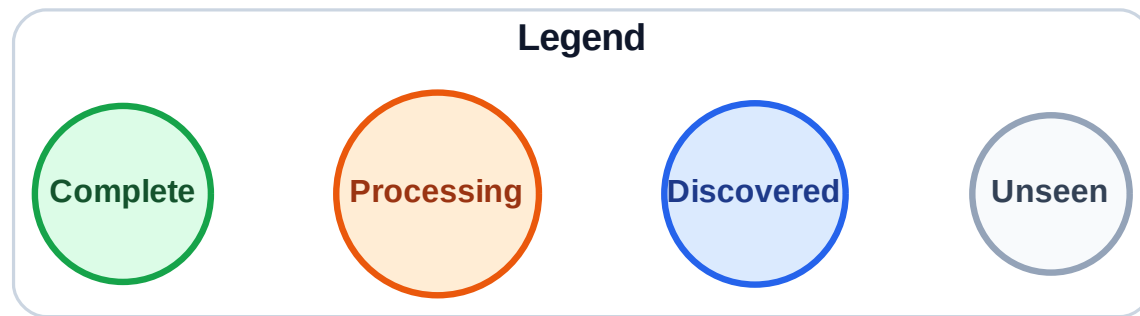
Queue: (empty)

Visited: (none)

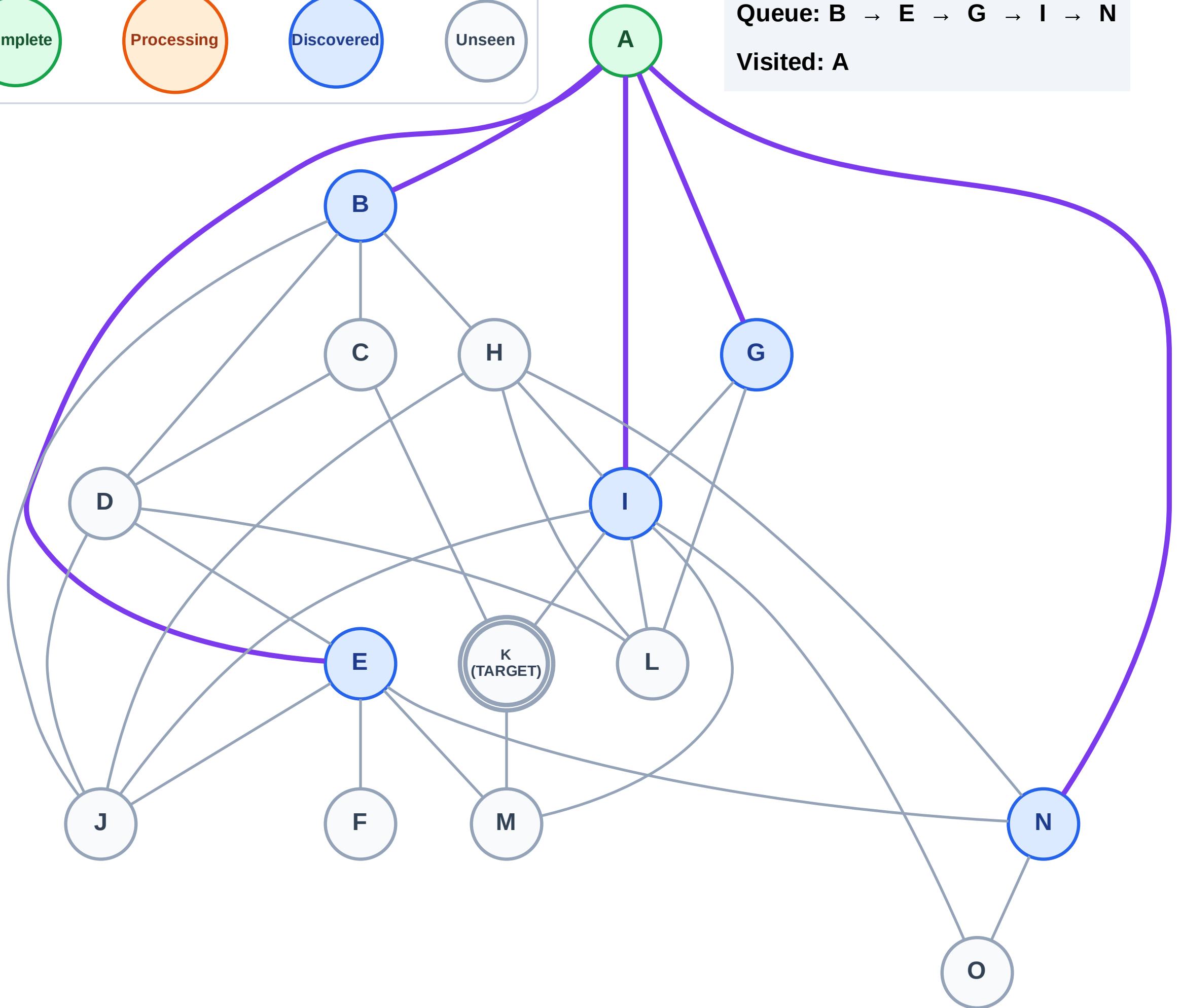


BFS Traversal

Step 3: Discover B, E, G, I, N from A



Queue: B → E → G → I → N
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

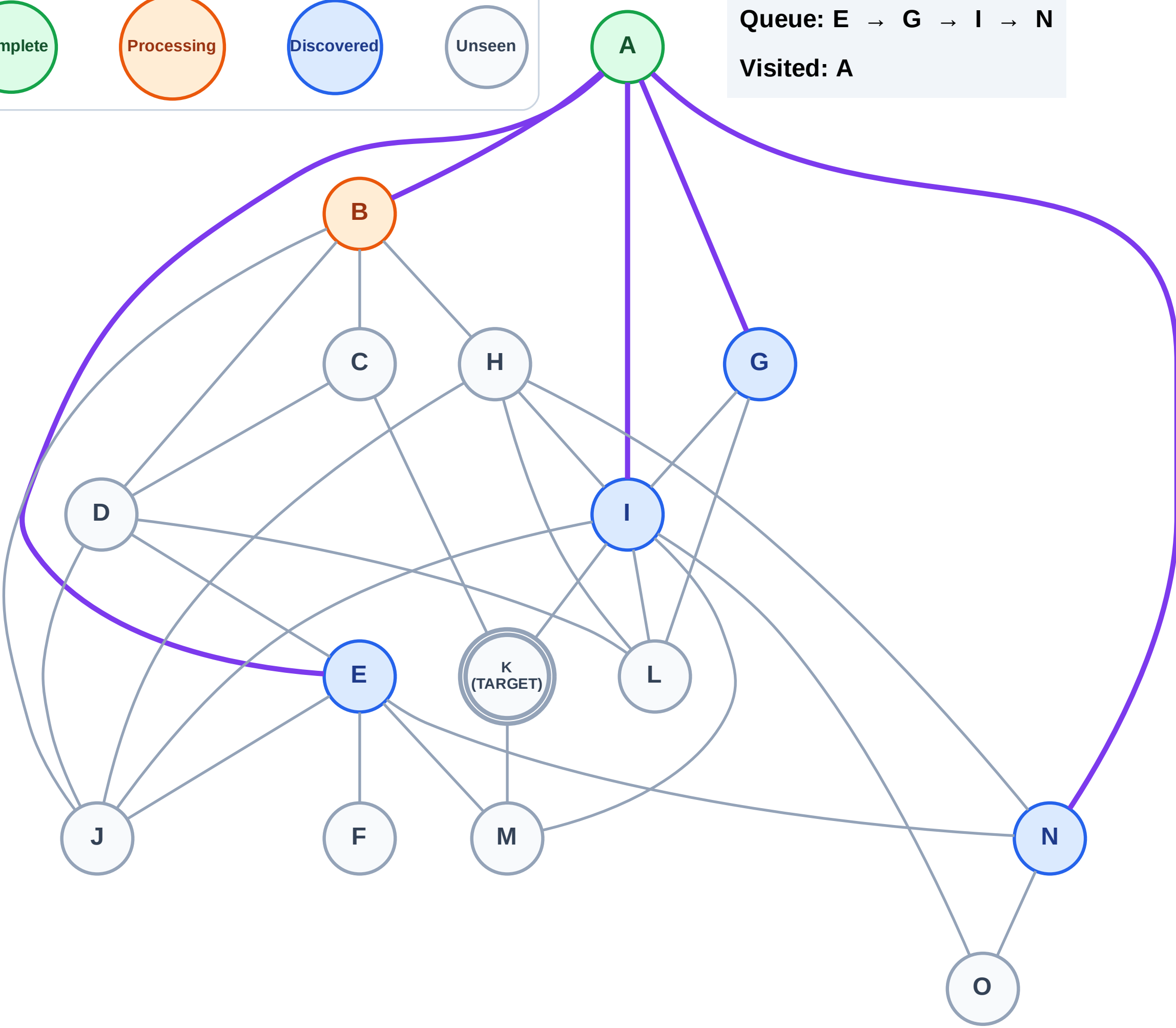
Processing

Discovered

Unseen

Queue: E → G → I → N

Visited: A



BFS Traversal

Step 5: Discover C, D, H, J from B

Legend

Complete

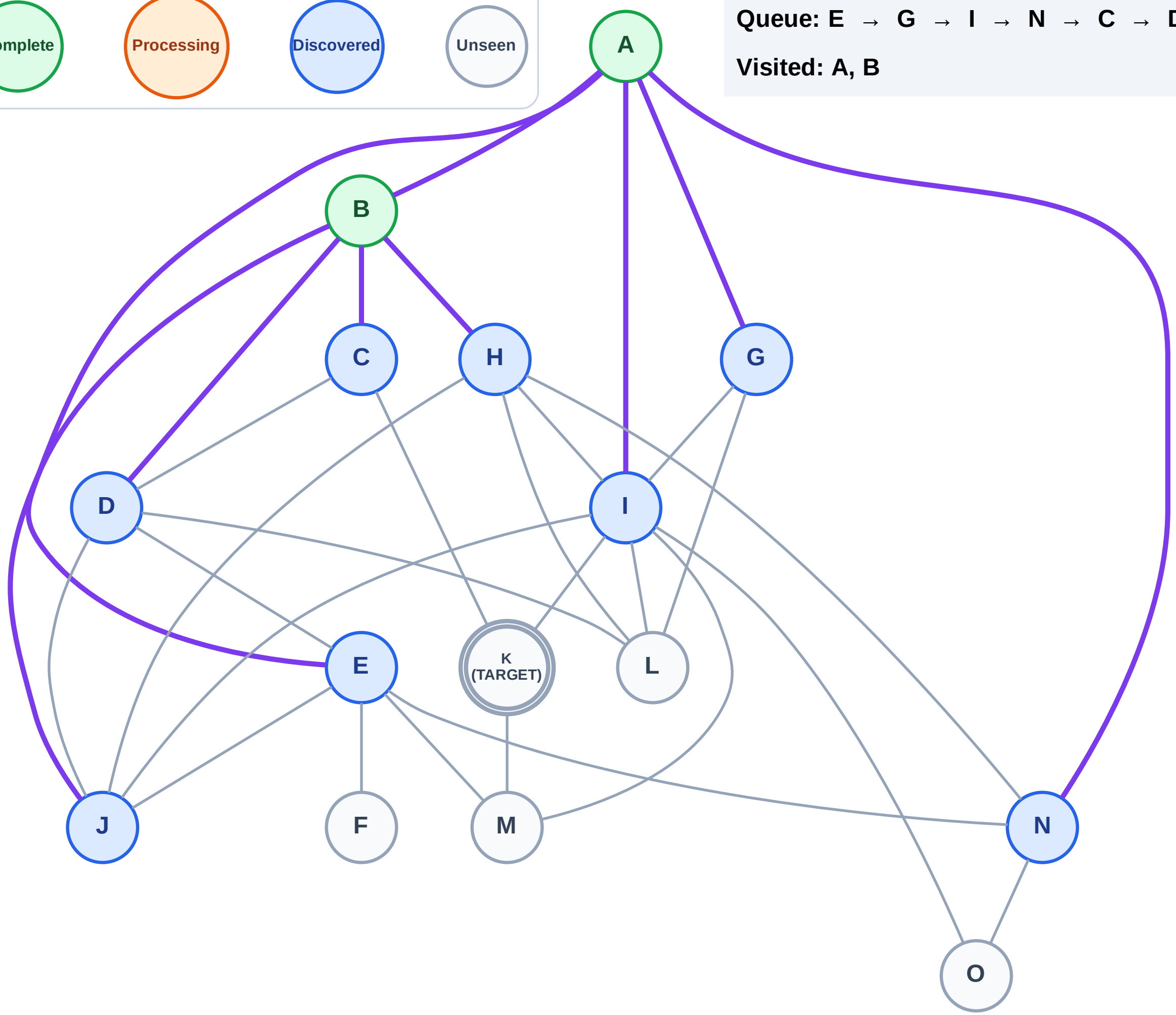
Processing

Discovered

Unseen

Queue: E → G → I → N → C → D → H → J

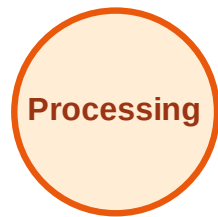
Visited: A, B



BFS Traversal

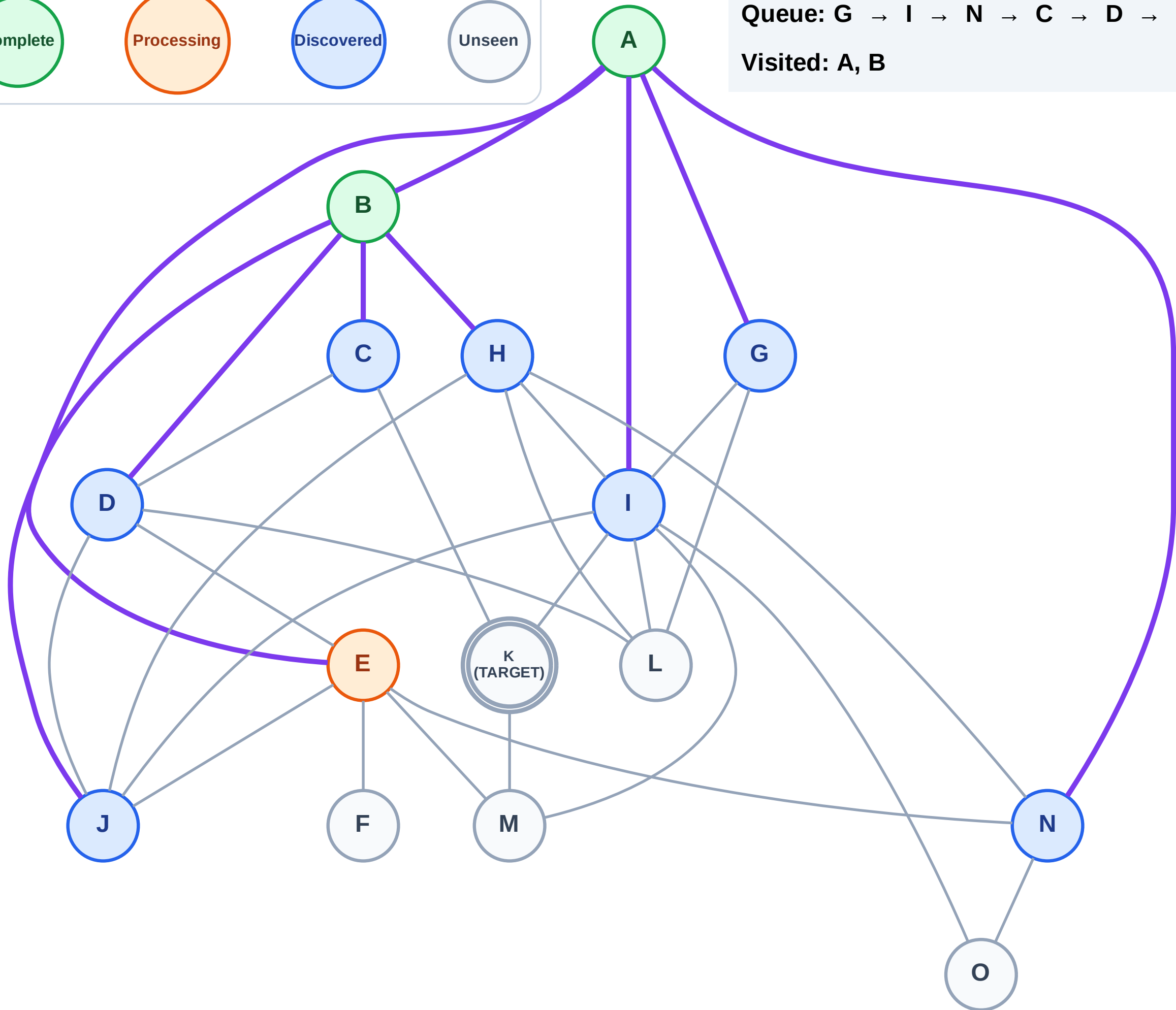
Step 6: Process node E

Legend



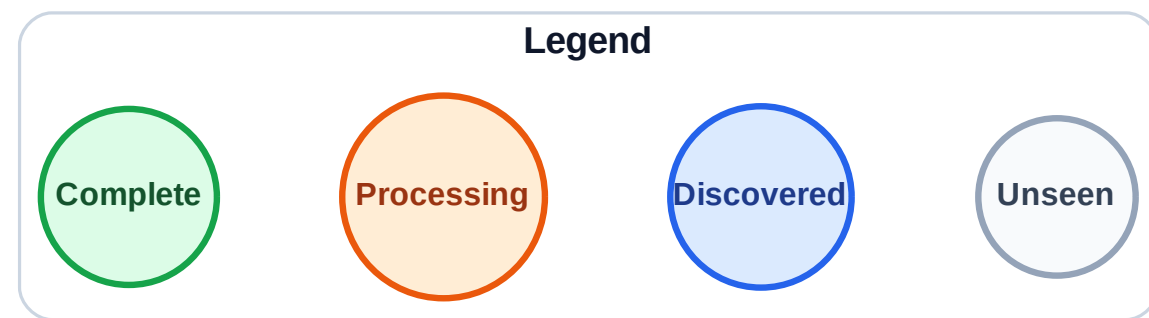
Queue: G → I → N → C → D → H → J

Visited: A, B

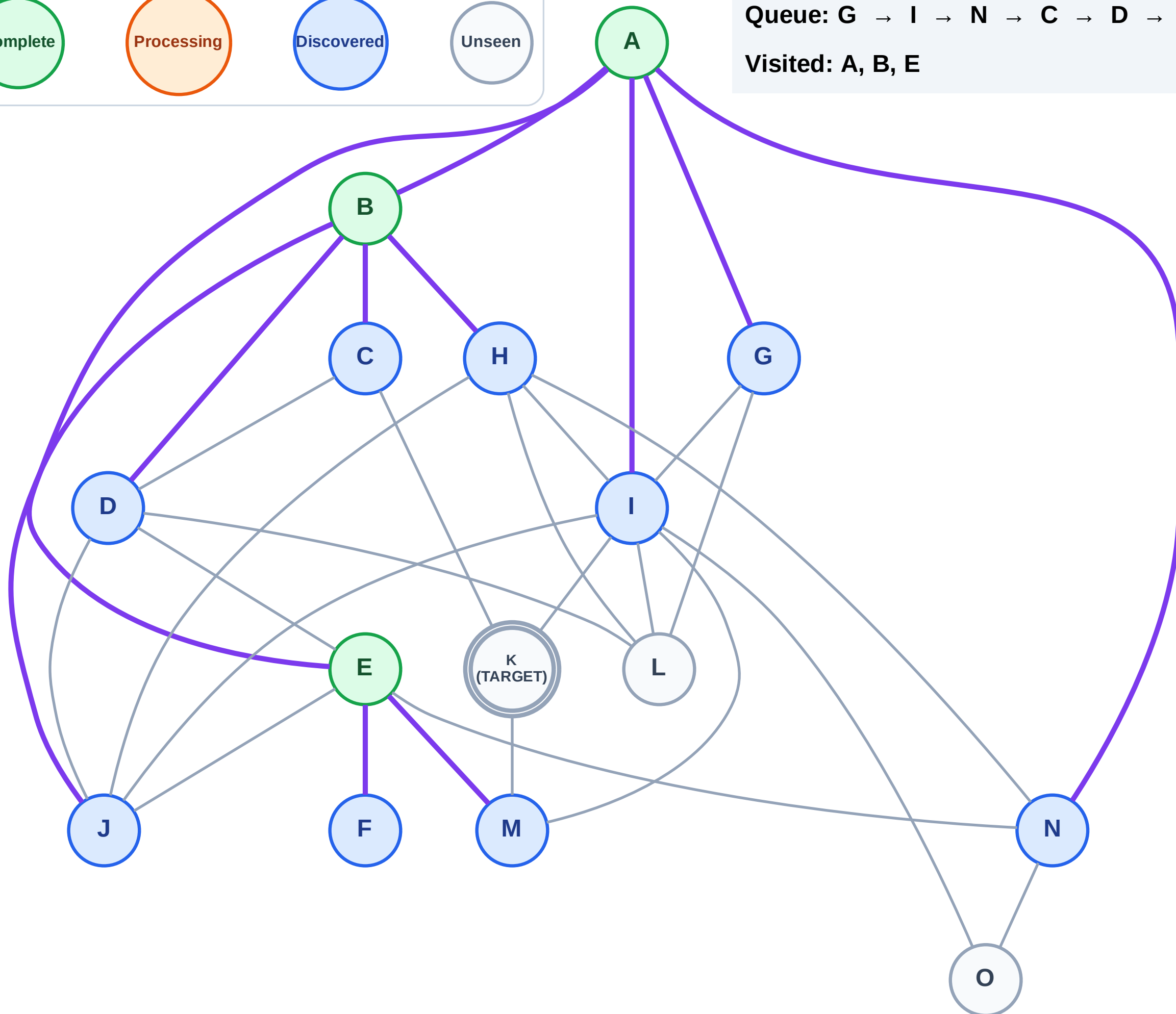


BFS Traversal

Step 7: Discover F, M from E



Queue: G → I → N → C → D → H → J → F → M
Visited: A, B, E



BFS Traversal

Step 8: Process node G

Legend

Complete

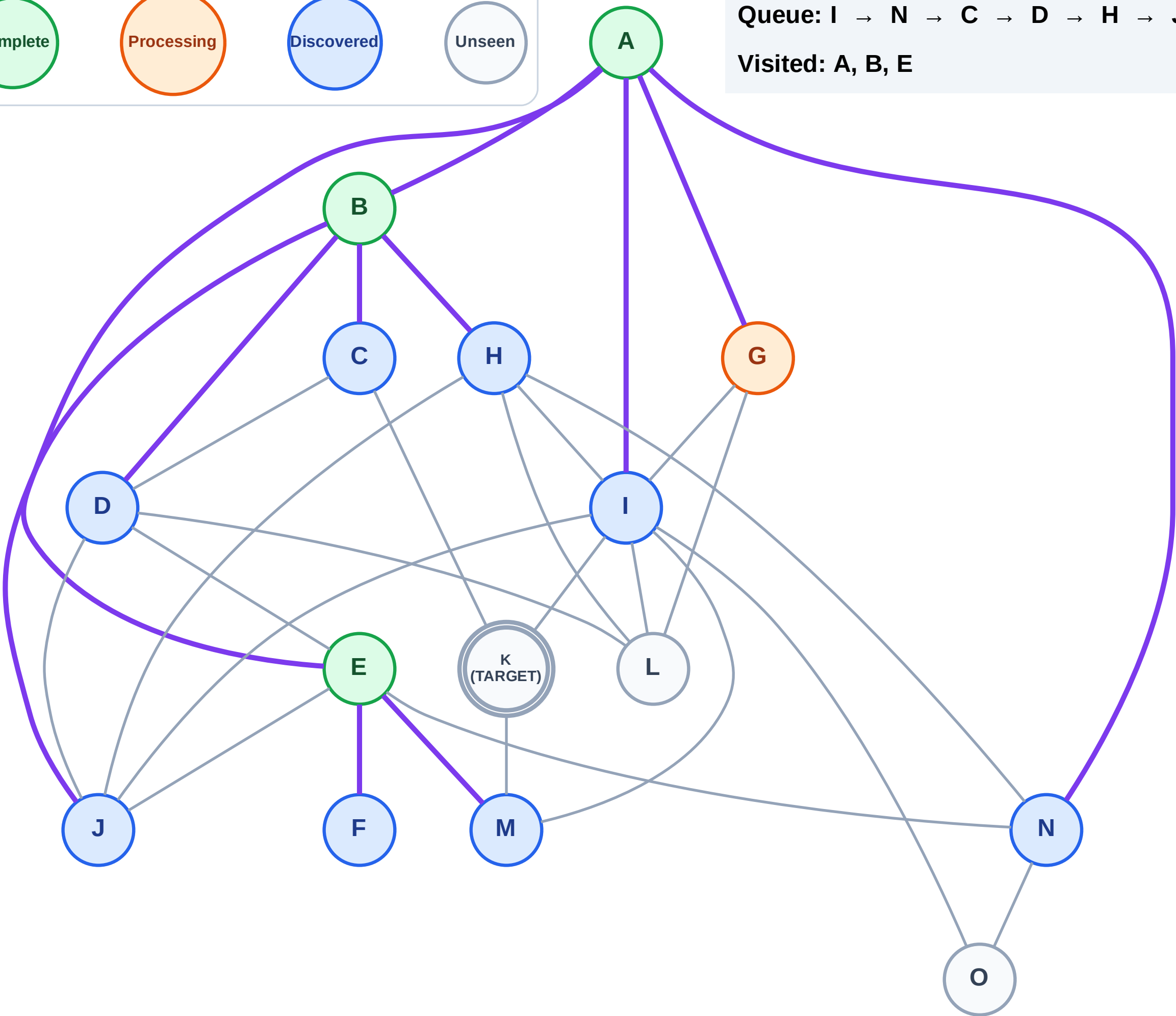
Processing

Discovered

Unseen

Queue: I → N → C → D → H → J → F → M

Visited: A, B, E



BFS Traversal

Step 9: Discover L from G

Legend



Complete



Processing



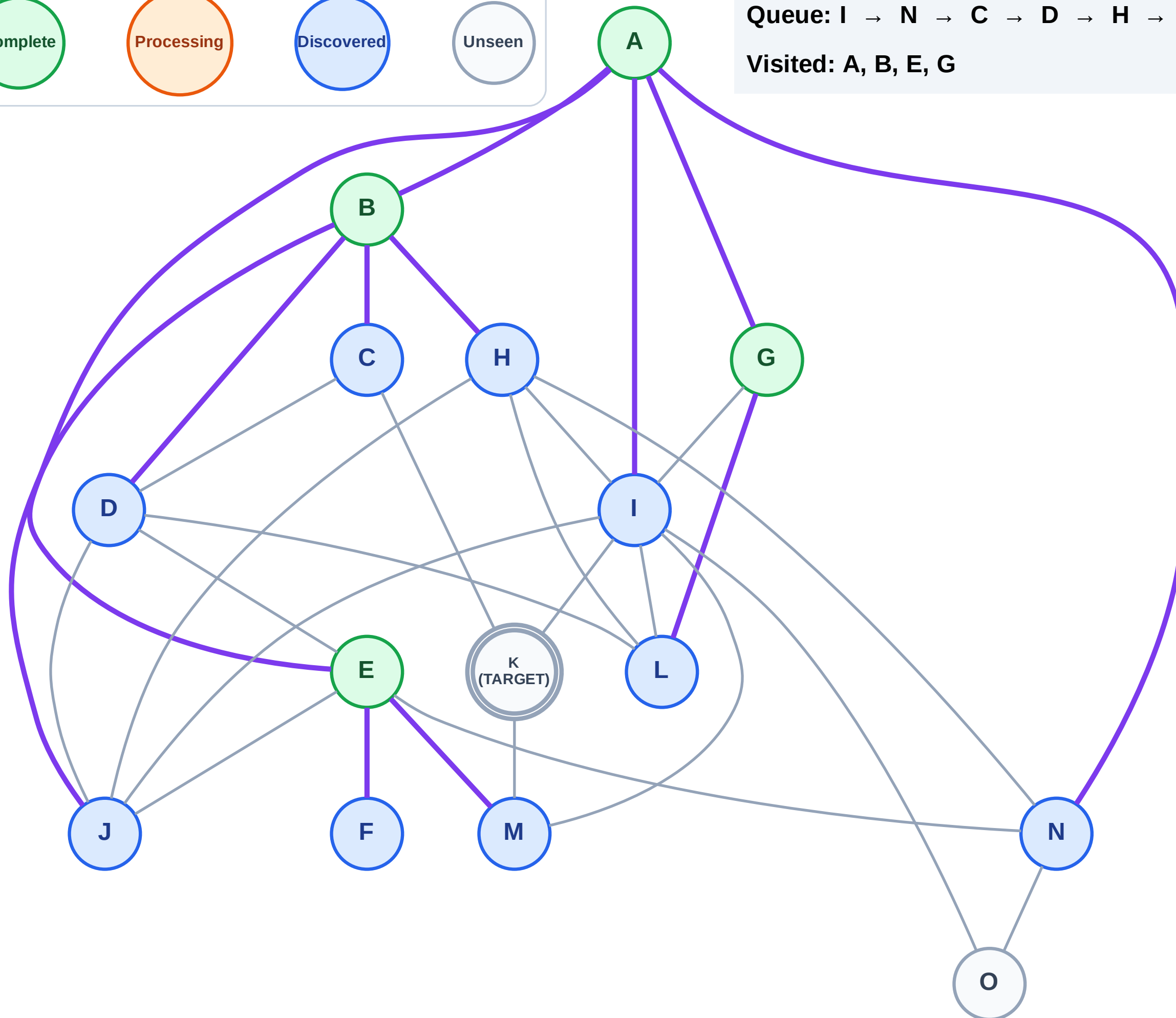
Discovered



Unseen

Queue: I → N → C → D → H → J → F → M → L

Visited: A, B, E, G



BFS Traversal

Step 10: Process node I

Legend

Complete

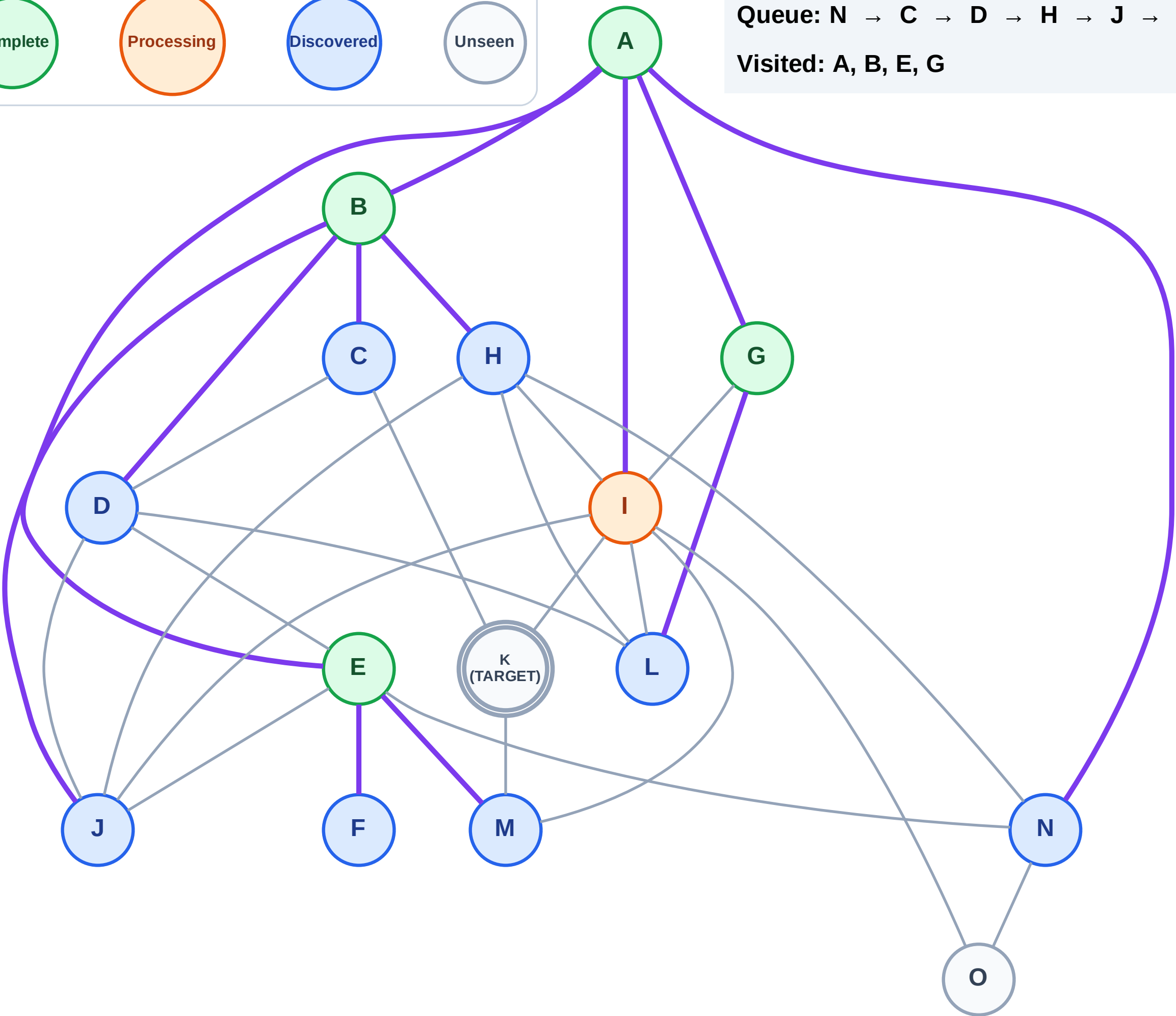
Processing

Discovered

Unseen

Queue: N → C → D → H → J → F → M → L

Visited: A, B, E, G



BFS Traversal

Step 11: Discover K, O from I

Legend

Complete

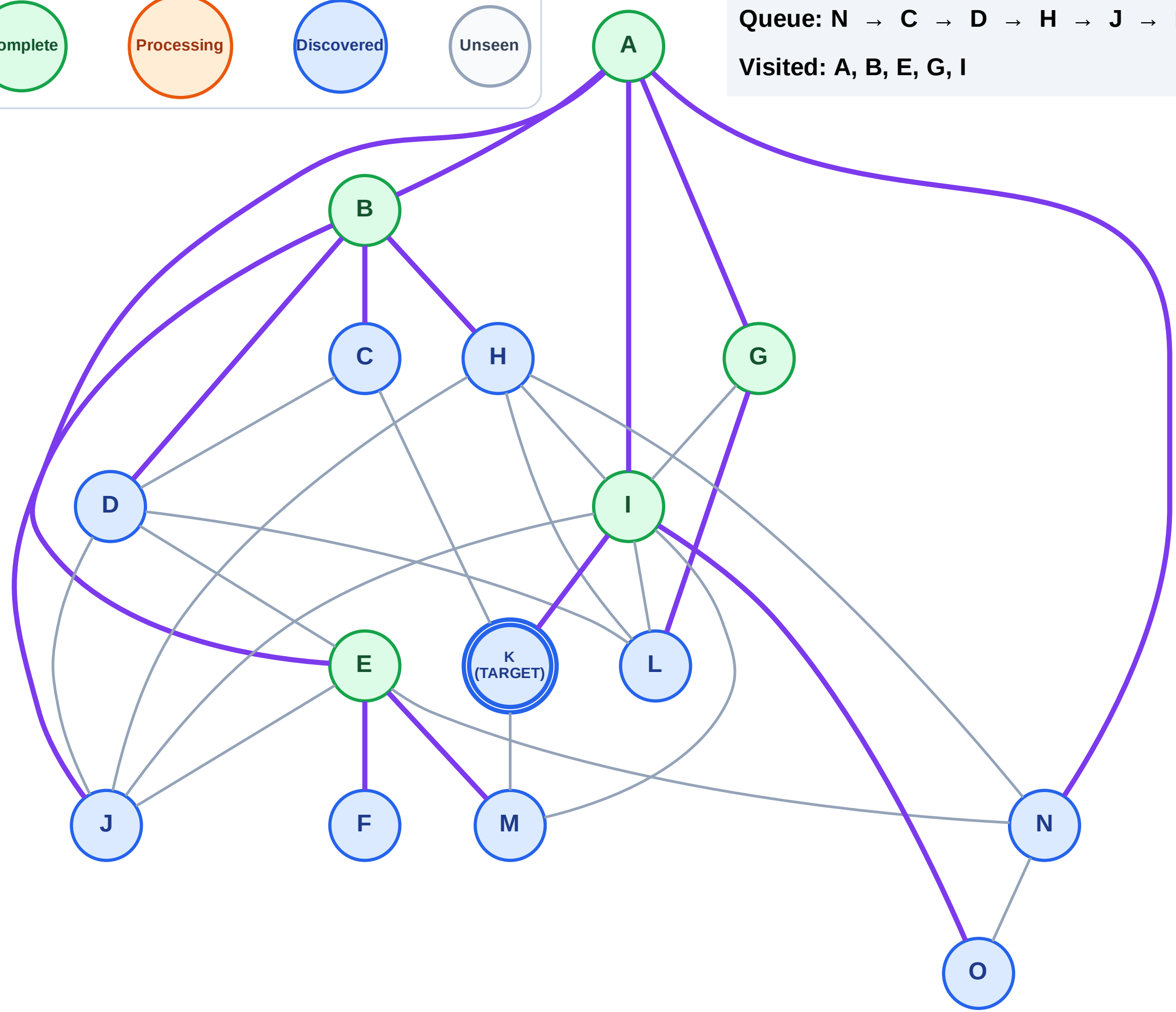
Processing

Discovered

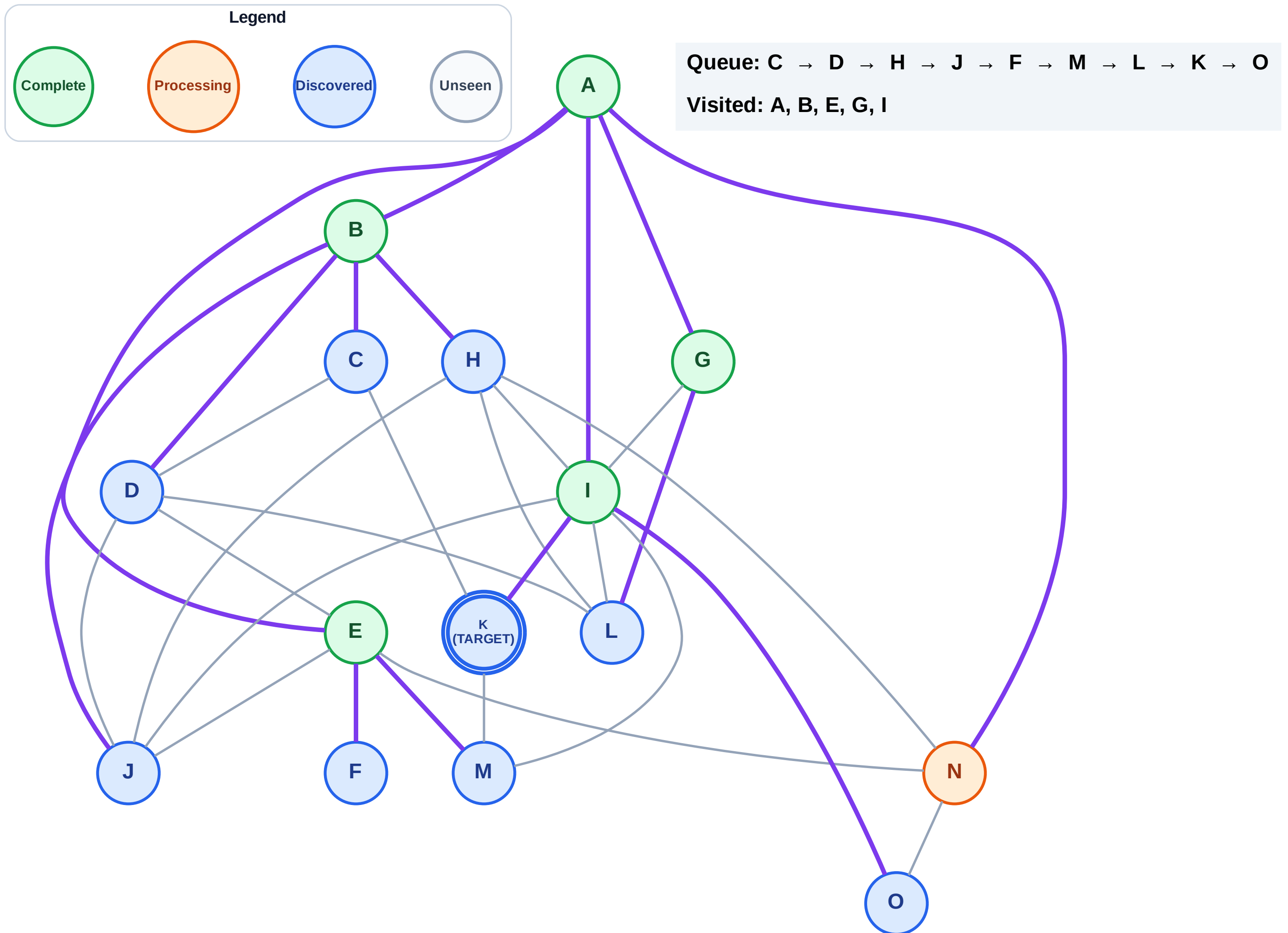
Unseen

Queue: N → C → D → H → J → F → M → L → K → O

Visited: A, B, E, G, I

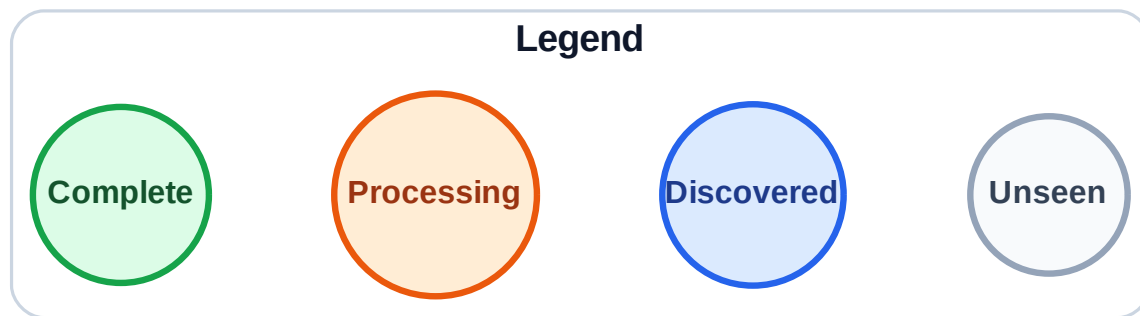


Step 12: Process node N



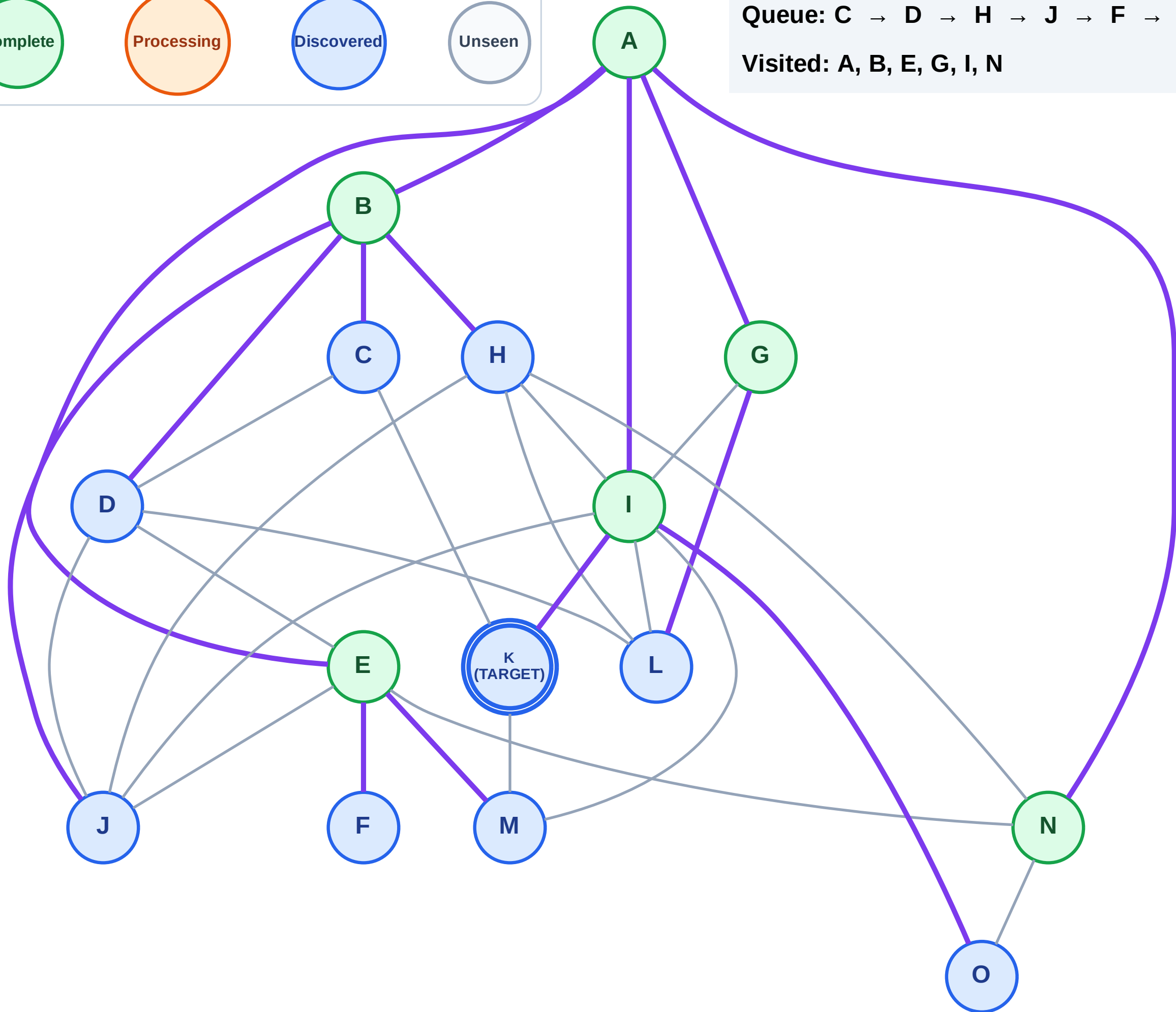
BFS Traversal

Step 13: No new neighbors from N



Queue: C → D → H → J → F → M → L → K → O

Visited: A, B, E, G, I, N



BFS Traversal

Step 14: Process node C

Legend

Complete

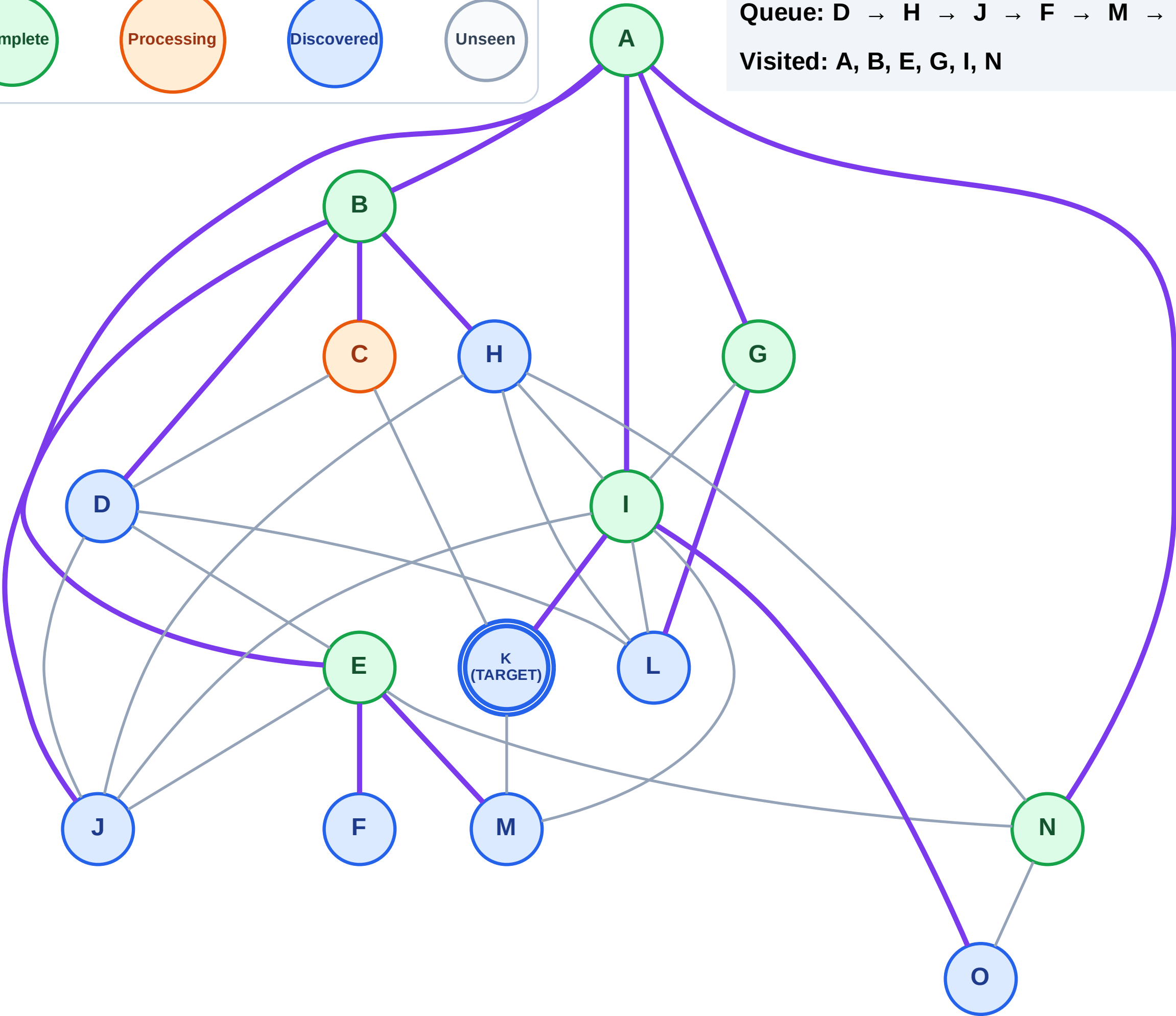
Processing

Discovered

Unseen

Queue: D → H → J → F → M → L → K → O

Visited: A, B, E, G, I, N



BFS Traversal

Step 15: No new neighbors from C

Legend

Complete

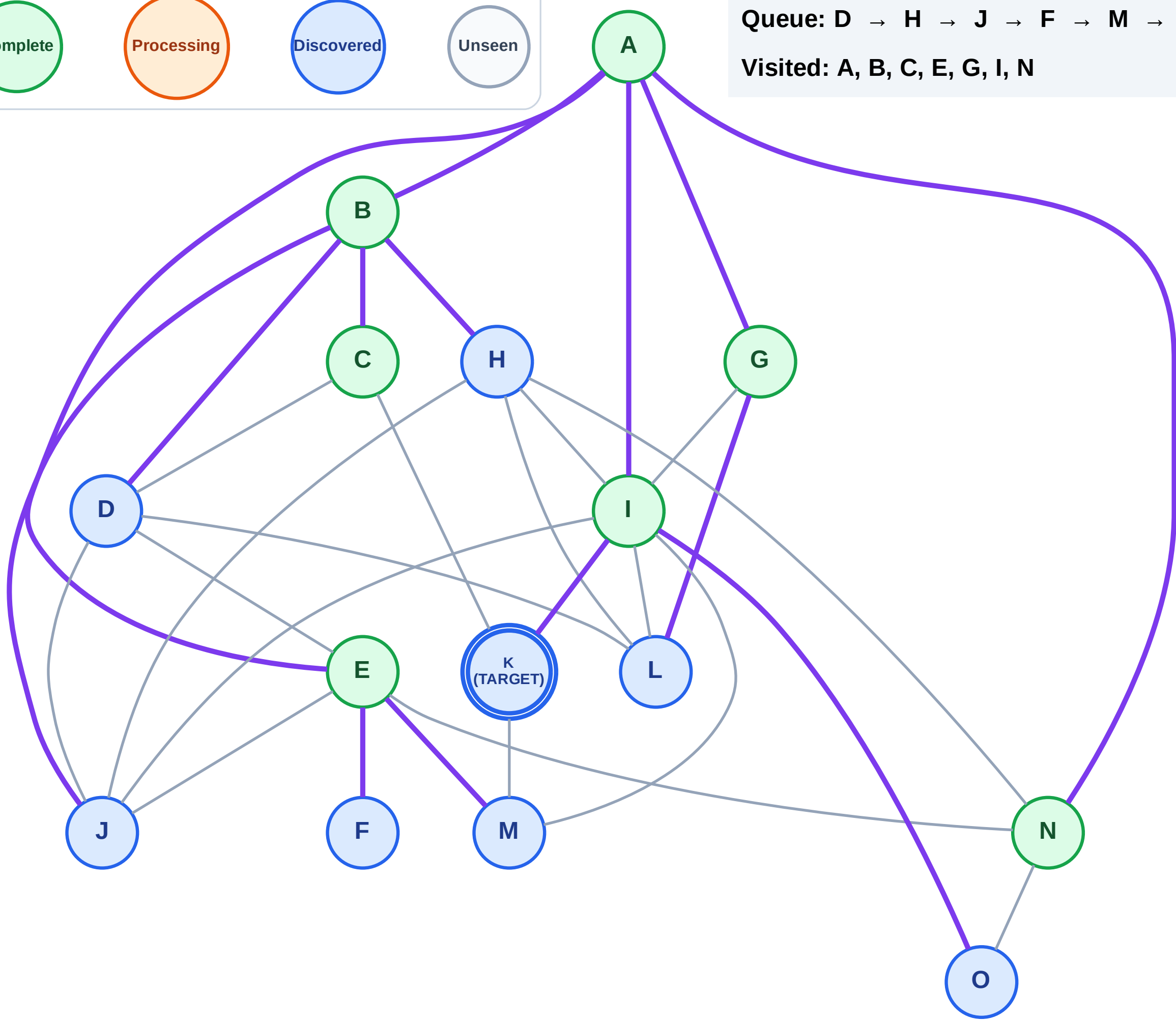
Processing

Discovered

Unseen

Queue: D → H → J → F → M → L → K → O

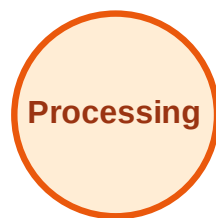
Visited: A, B, C, E, G, I, N



BFS Traversal

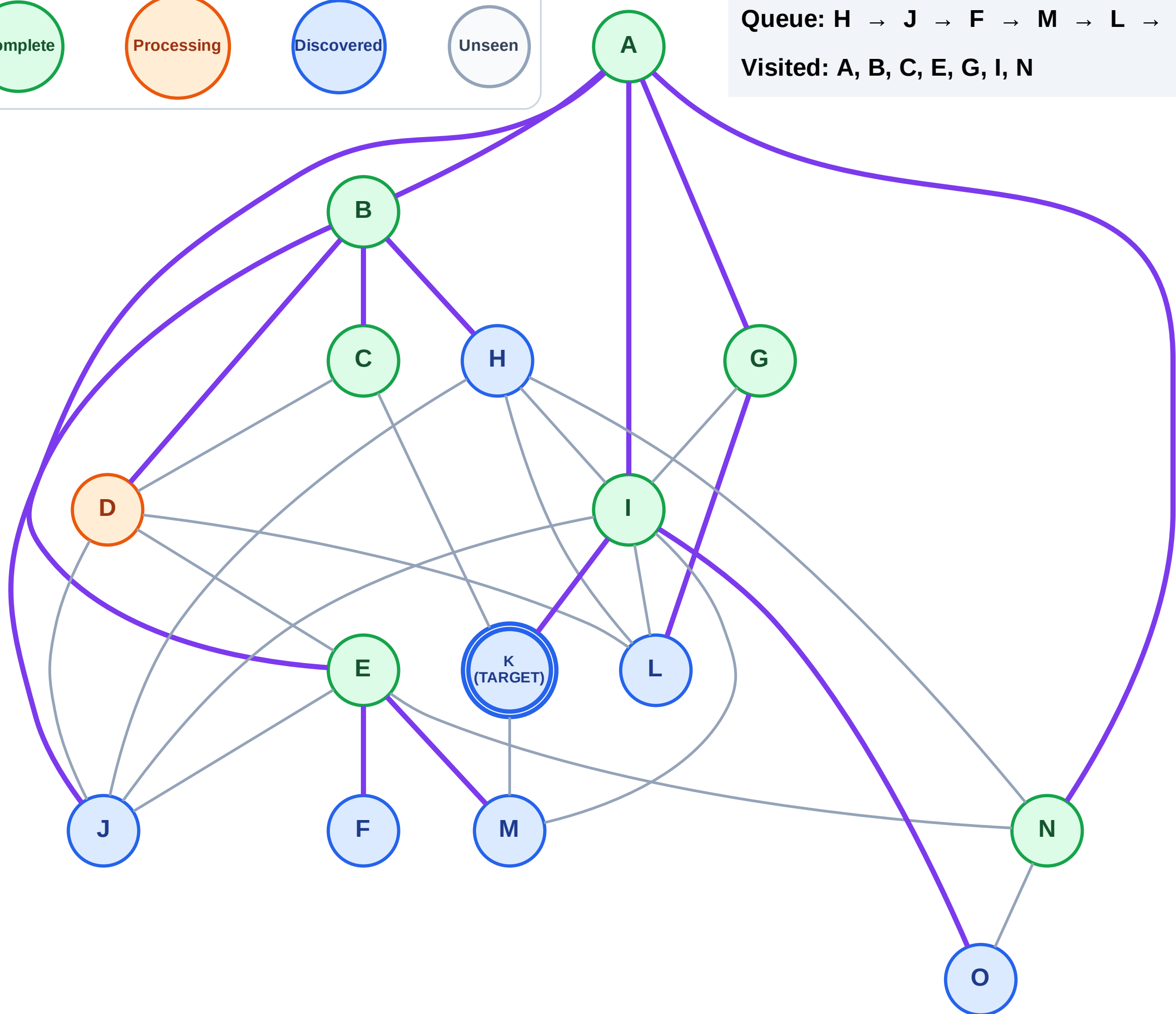
Step 16: Process node D

Legend



Queue: H → J → F → M → L → K → O

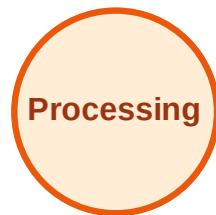
Visited: A, B, C, E, G, I, N



BFS Traversal

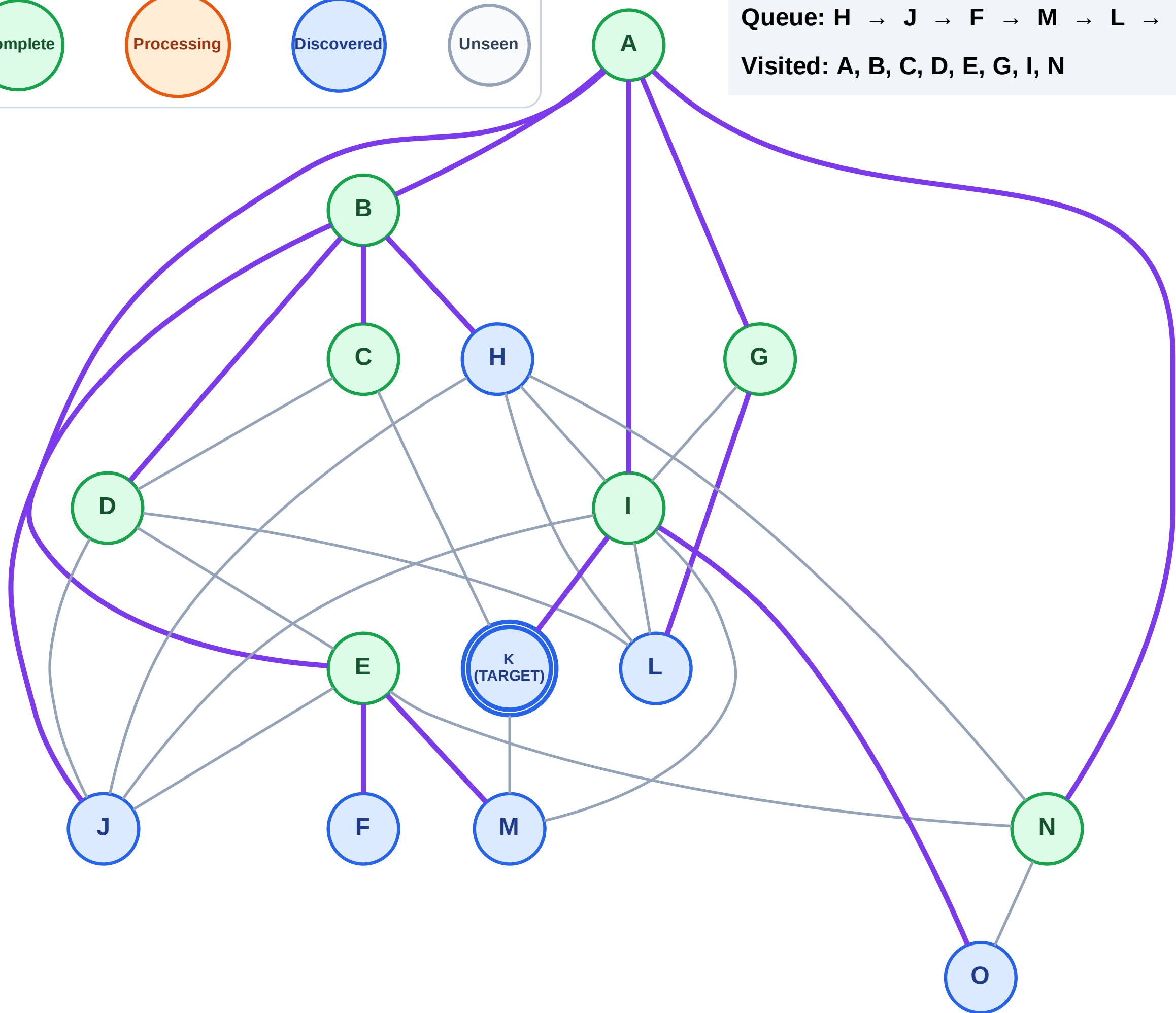
Step 17: No new neighbors from D

Legend



Queue: H → J → F → M → L → K → O

Visited: A, B, C, D, E, G, I, N



BFS Traversal

Step 18: Process node H

Legend

Complete

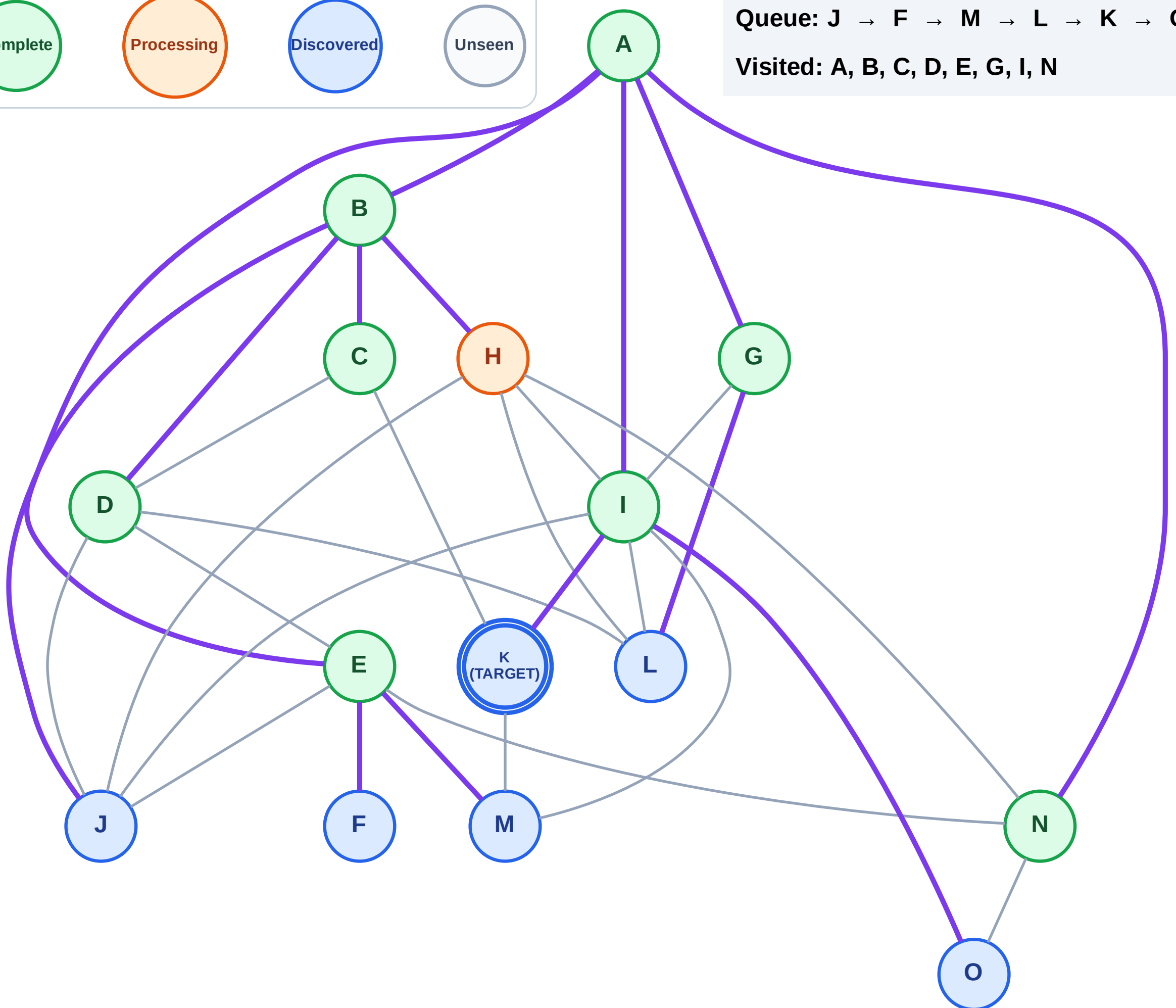
Processing

Discovered

Unseen

Queue: J → F → M → L → K → O

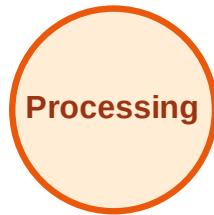
Visited: A, B, C, D, E, G, I, N



BFS Traversal

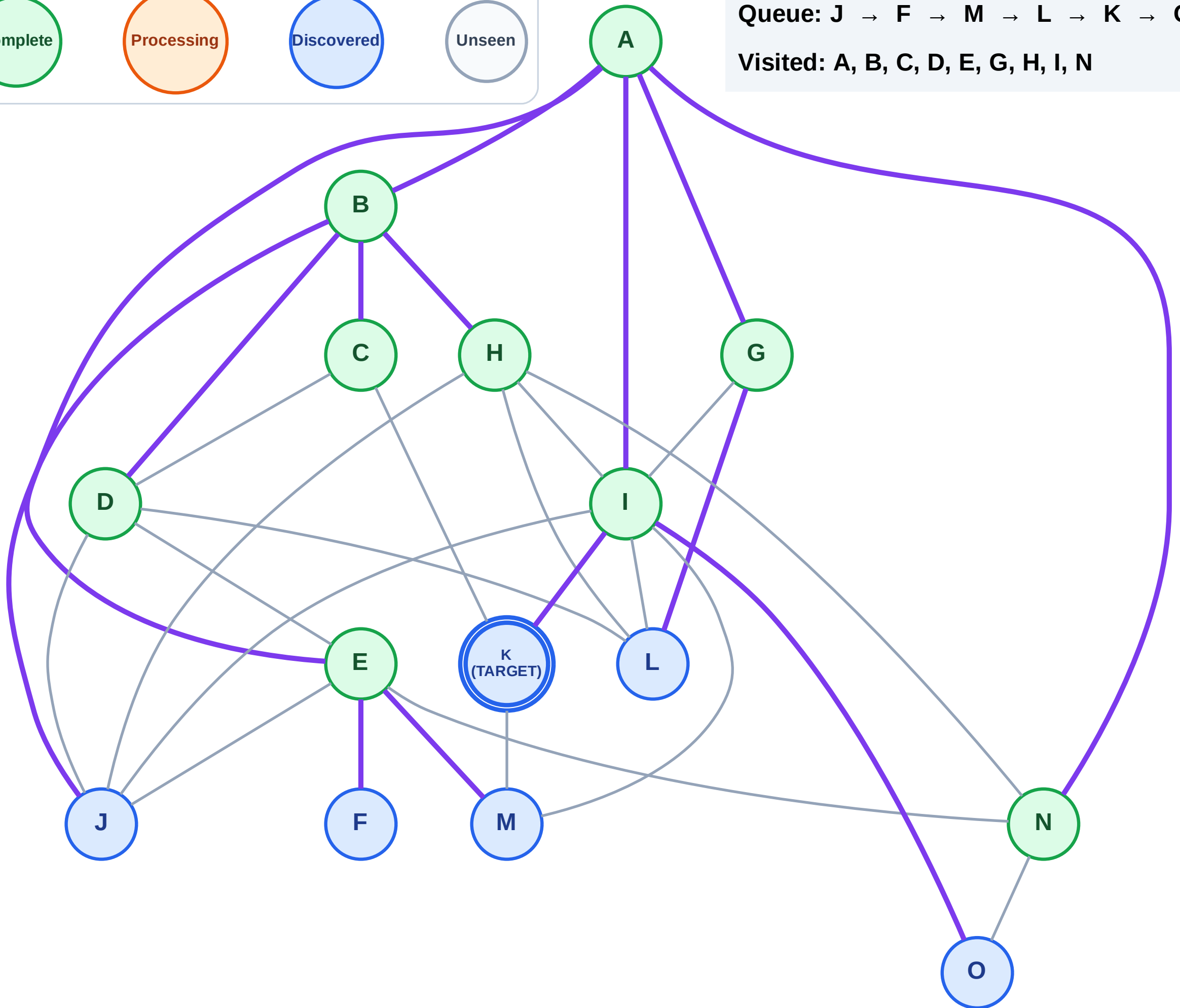
Step 19: No new neighbors from H

Legend



Queue: J → F → M → L → K → O

Visited: A, B, C, D, E, G, H, I, N



BFS Traversal

Step 20: Process node J

Legend

Complete

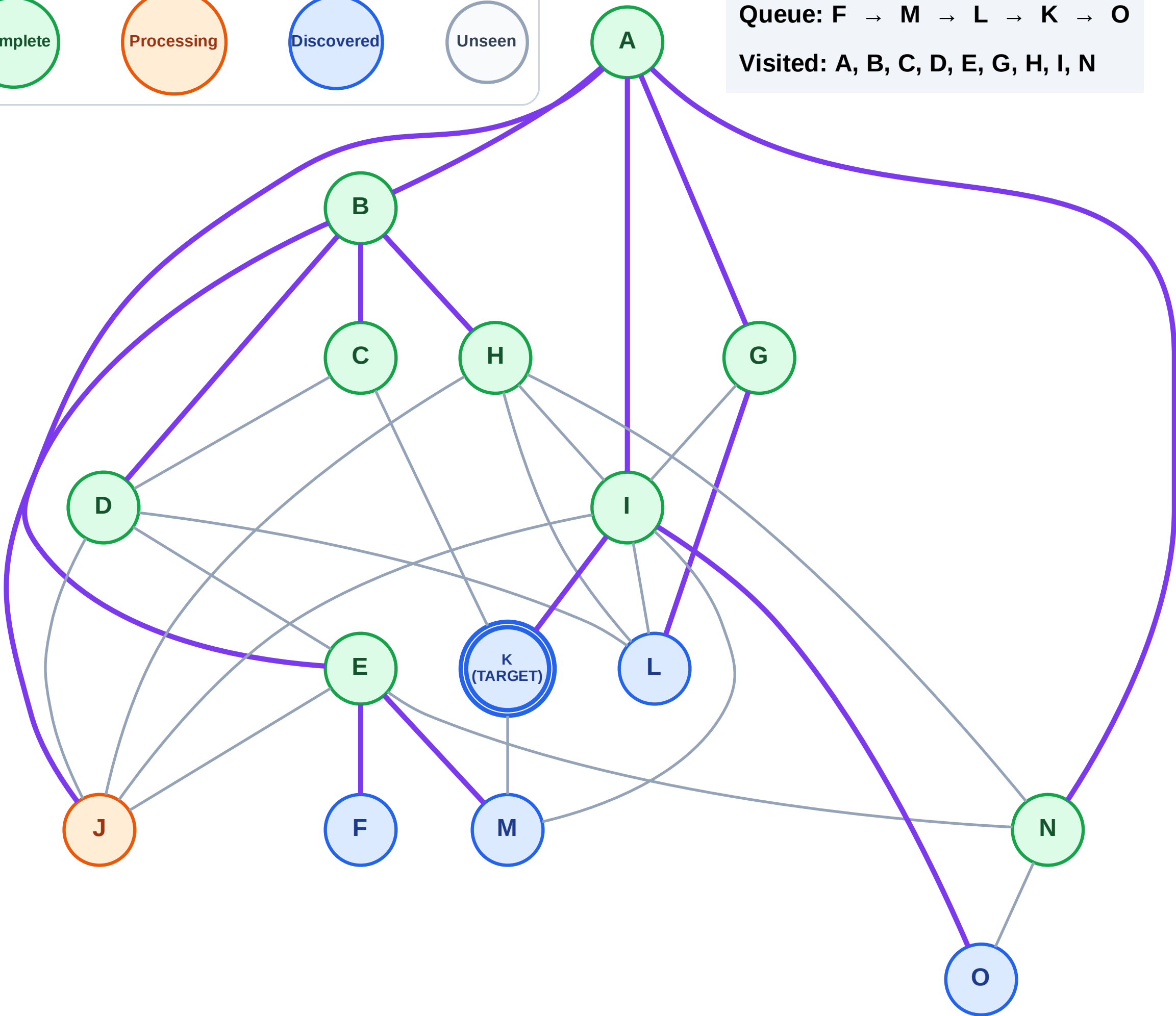
Processing

Discovered

Unseen

Queue: F → M → L → K → O

Visited: A, B, C, D, E, G, H, I, N



BFS Traversal

Step 21: No new neighbors from J

Legend

Complete

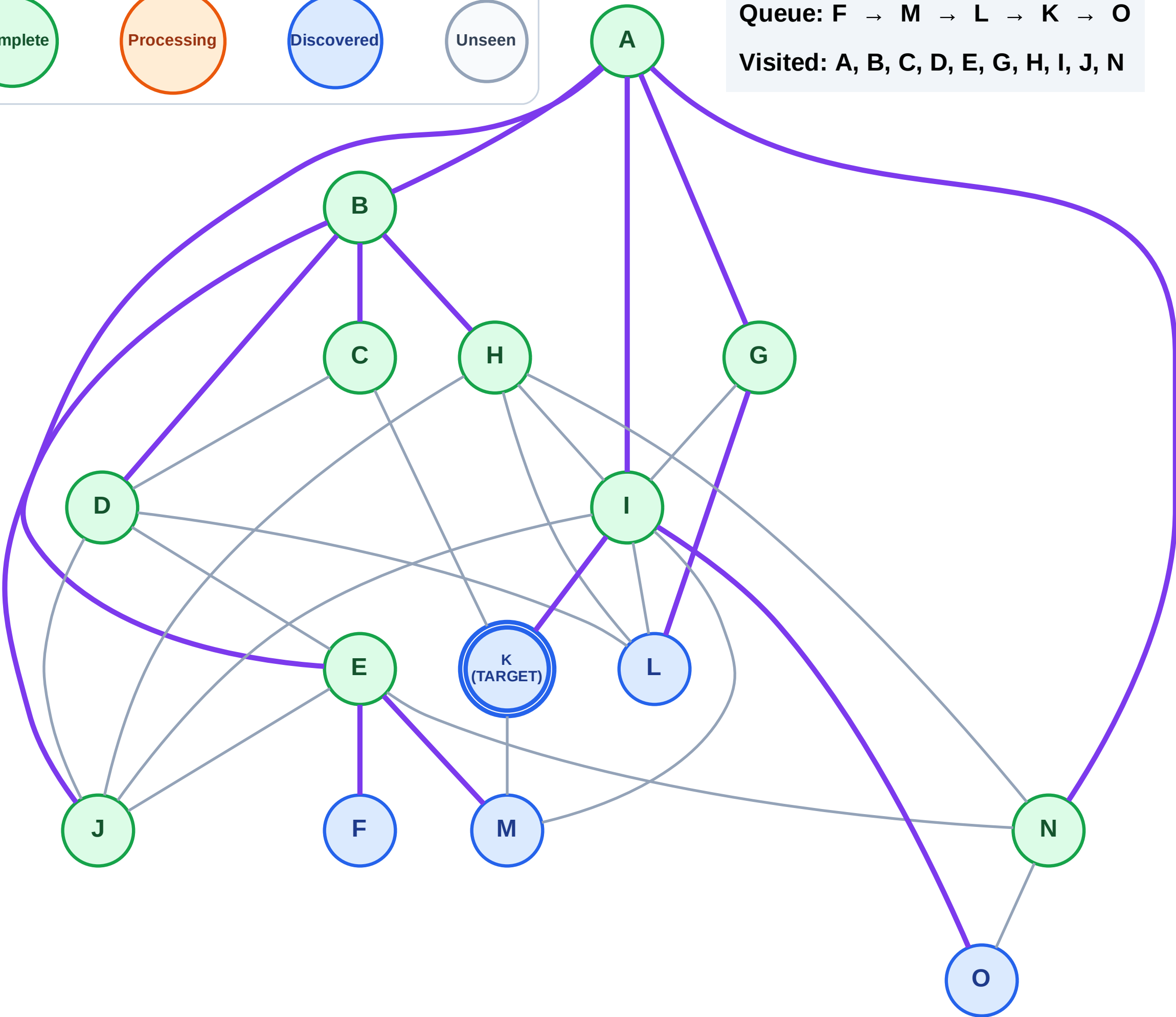
Processing

Discovered

Unseen

Queue: F → M → L → K → O

Visited: A, B, C, D, E, G, H, I, J, N



BFS Traversal

Step 22: Process node F

Legend

Complete

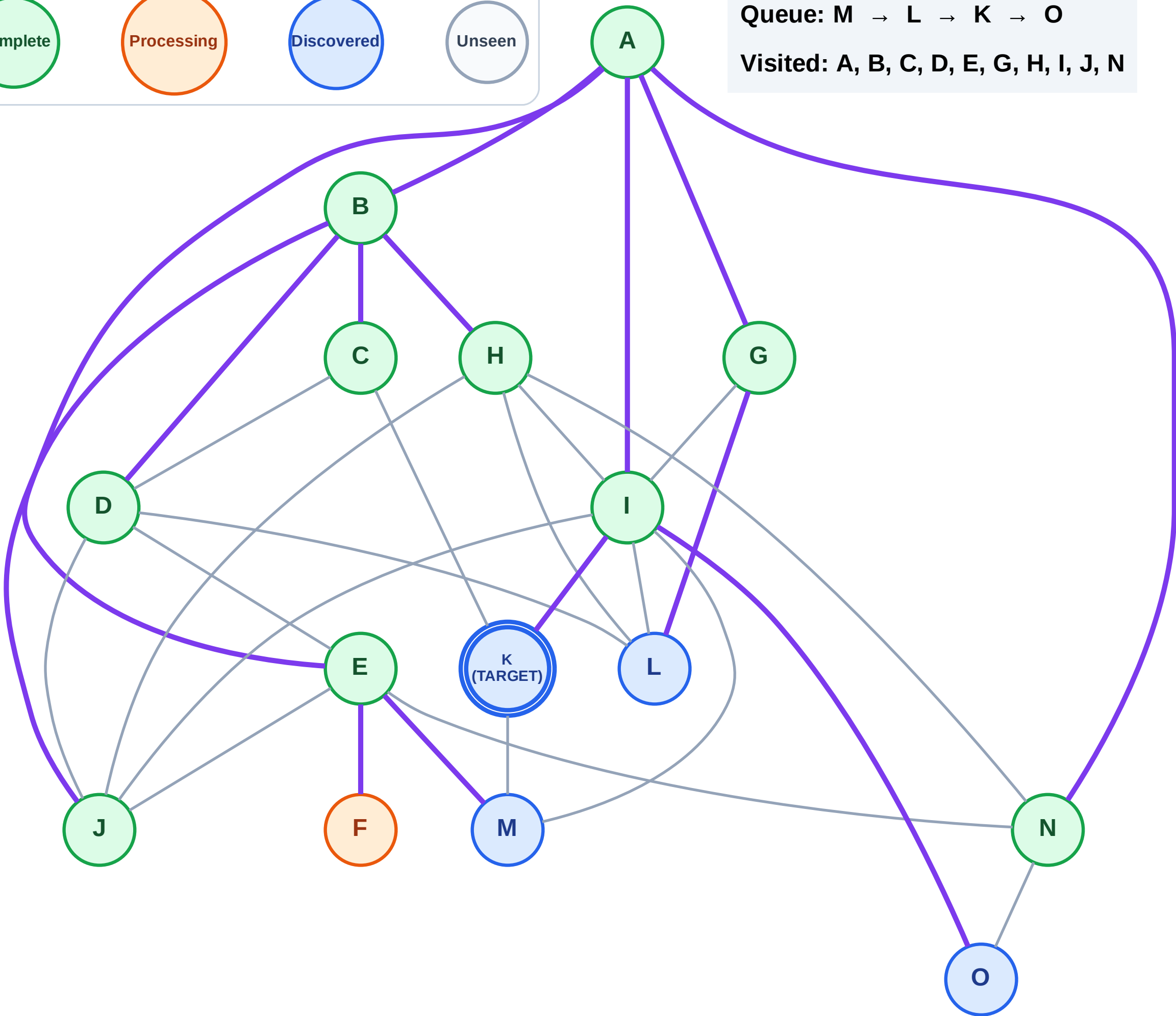
Processing

Discovered

Unseen

Queue: M → L → K → O

Visited: A, B, C, D, E, G, H, I, J, N



BFS Traversal

Step 23: No new neighbors from F

Legend

Complete

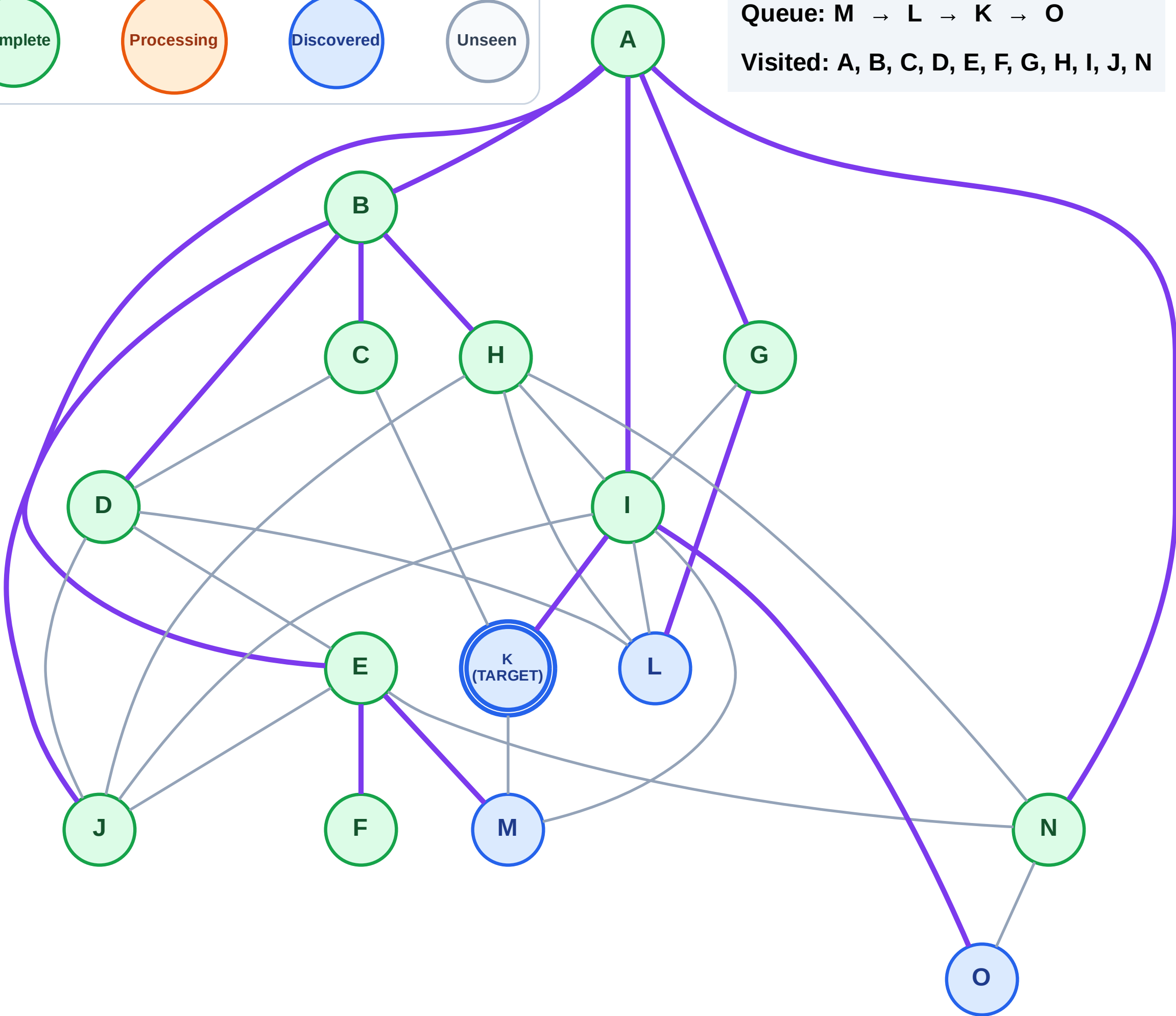
Processing

Discovered

Unseen

Queue: M → L → K → O

Visited: A, B, C, D, E, F, G, H, I, J, N



BFS Traversal

Step 24: Process node M

Legend

Complete

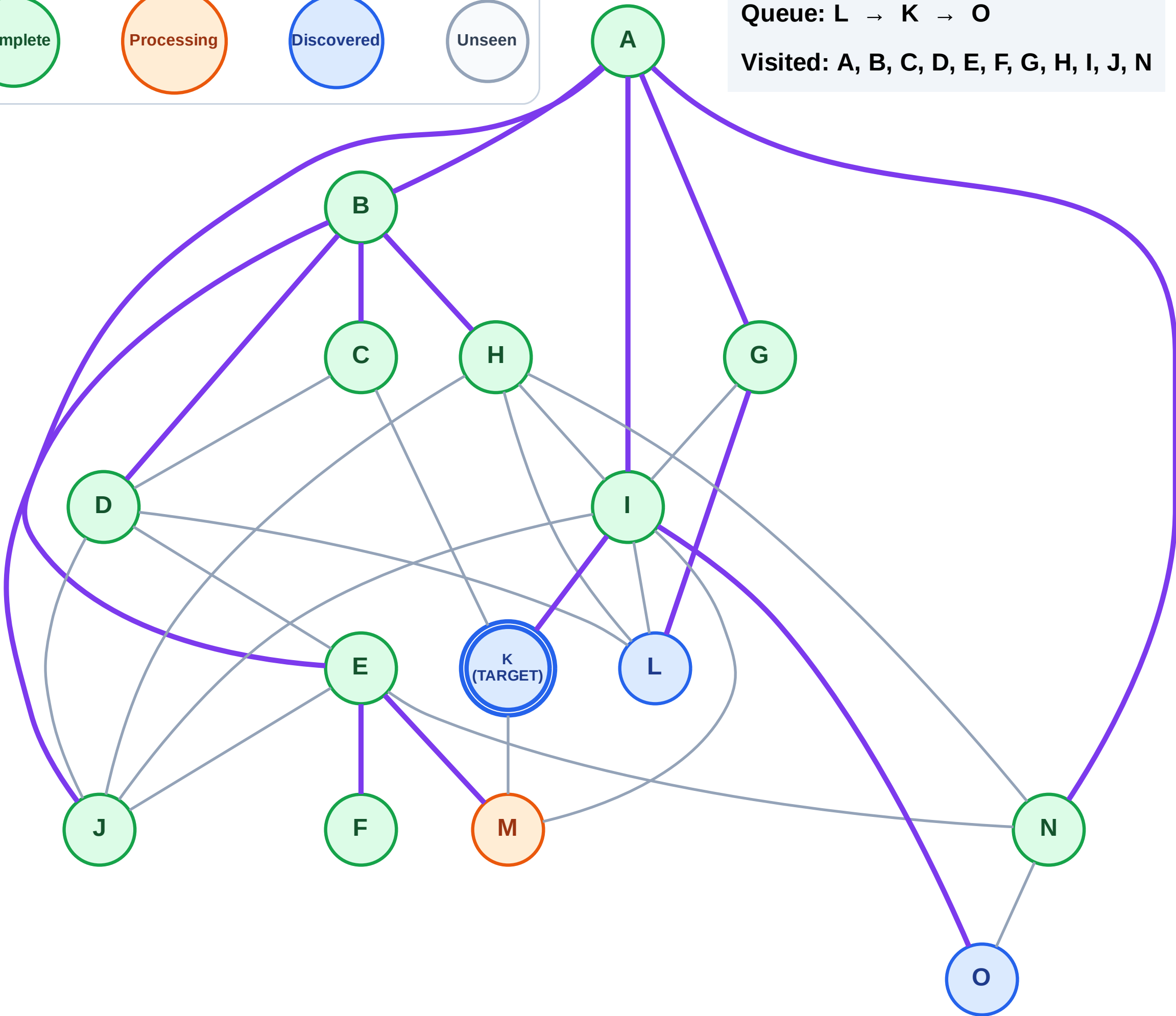
Processing

Discovered

Unseen

Queue: L → K → O

Visited: A, B, C, D, E, F, G, H, I, J, N

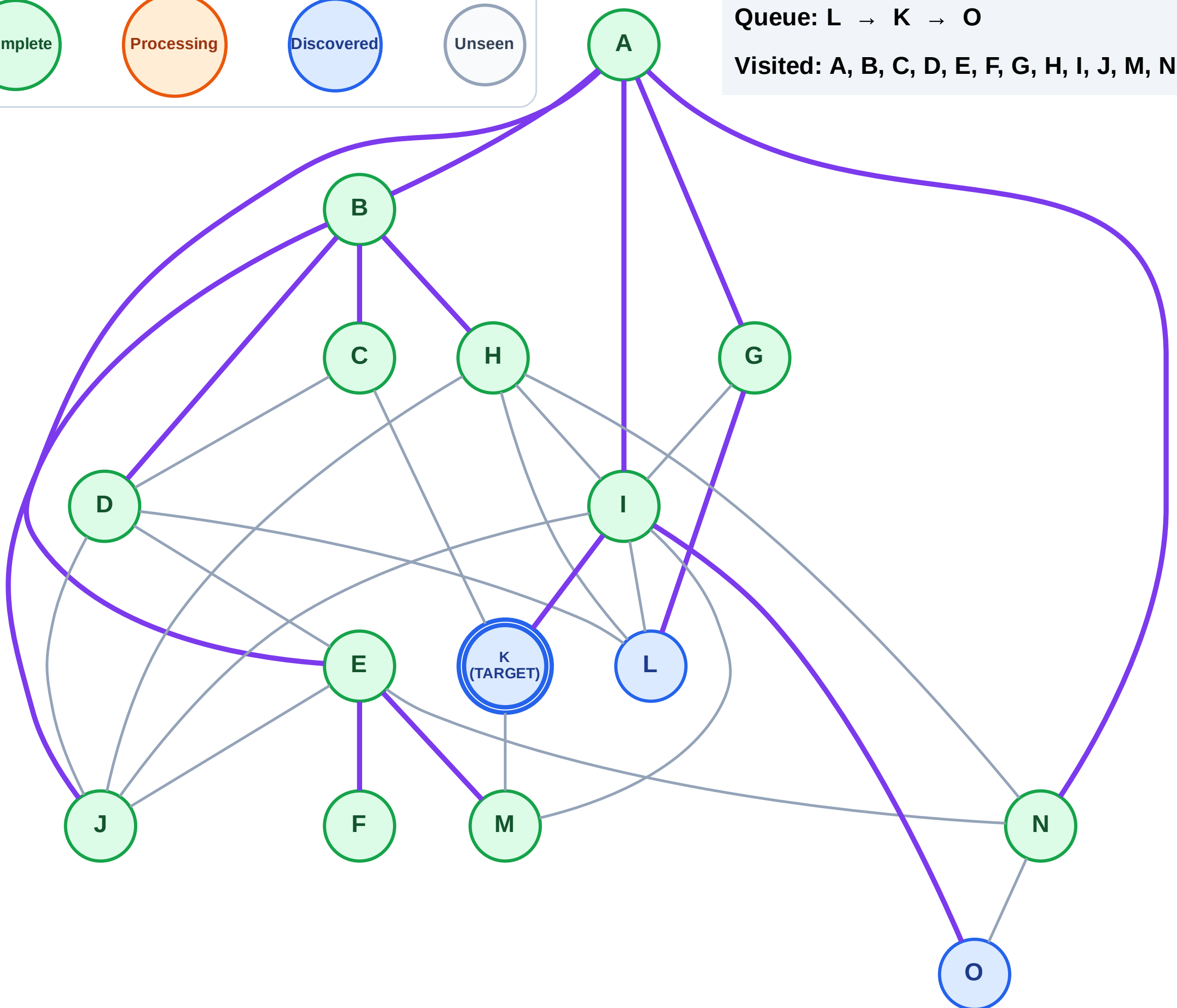


Step 25: No new neighbors from M

Legend

Unseen

Visited: A, B, C, D, E, F, G, H, I, J, M, N



BFS Traversal

Step 26: Process node L

Legend

Complete

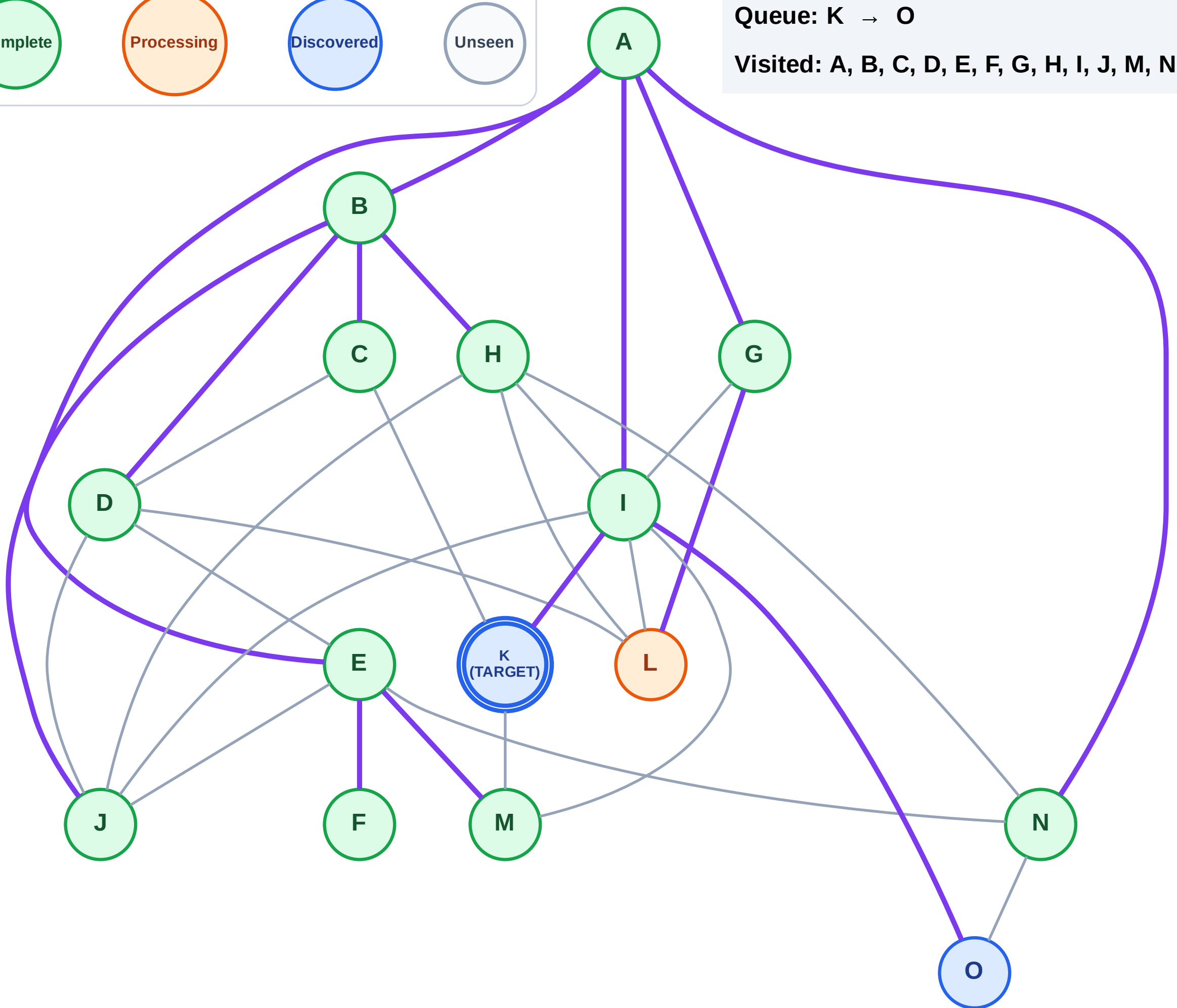
Processing

Discovered

Unseen

Queue: K → O

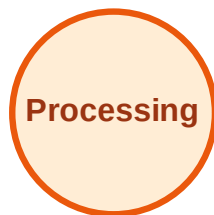
Visited: A, B, C, D, E, F, G, H, I, J, M, N



BFS Traversal

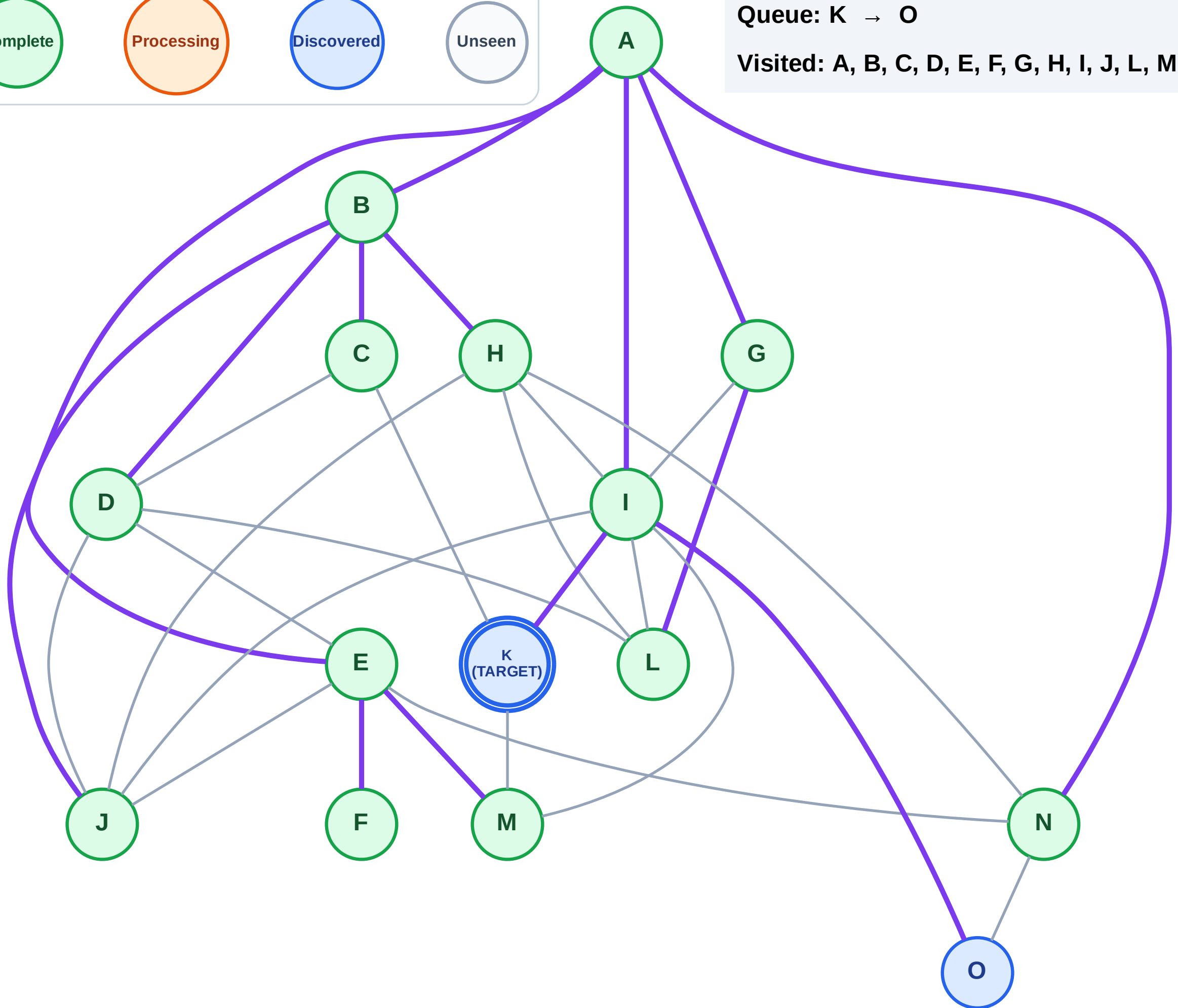
Step 27: No new neighbors from L

Legend



Queue: K → O

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 28: Process node K

Legend

Complete

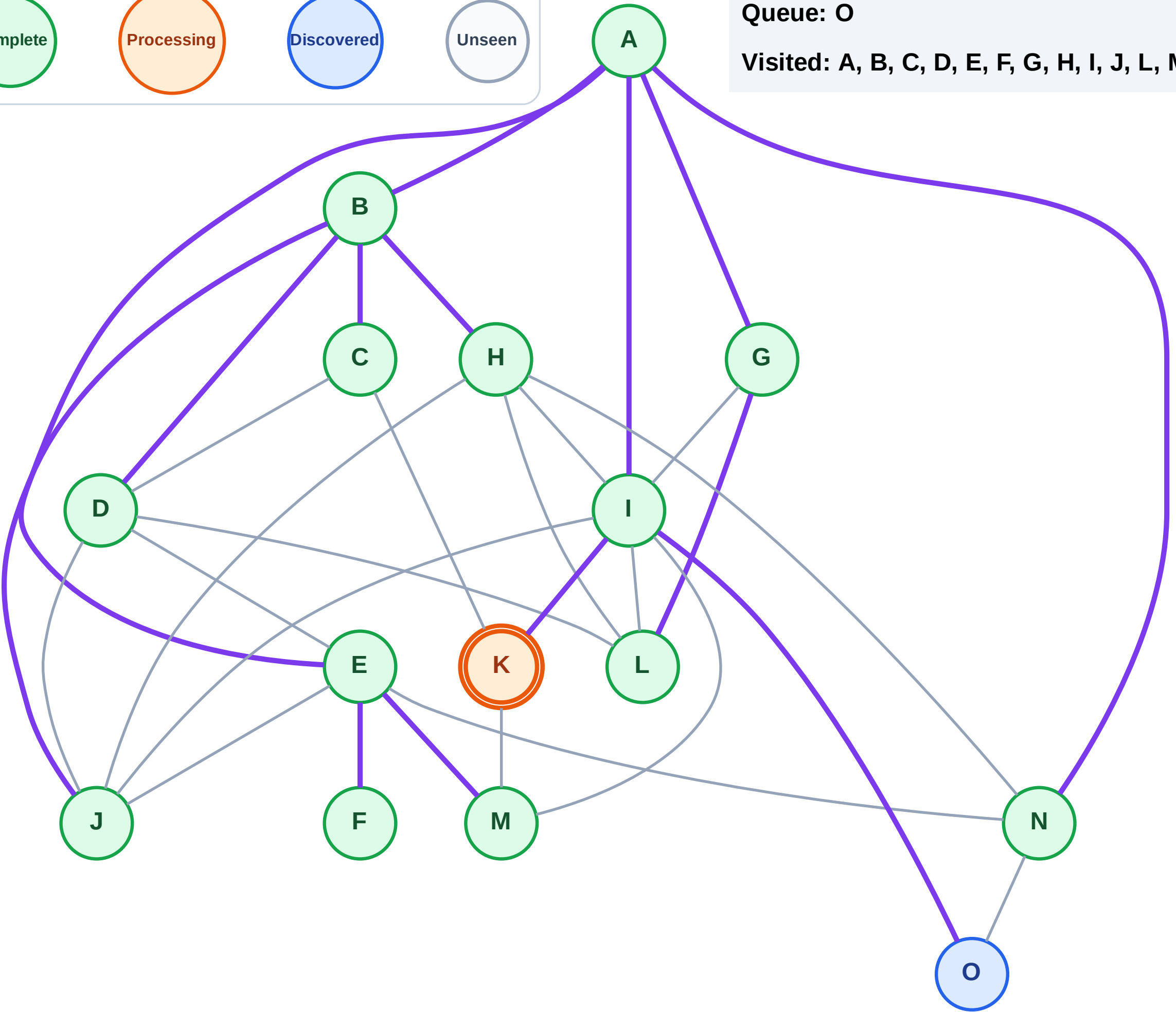
Processing

Discovered

Unseen

Queue: O

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → I → K

Legend

Complete

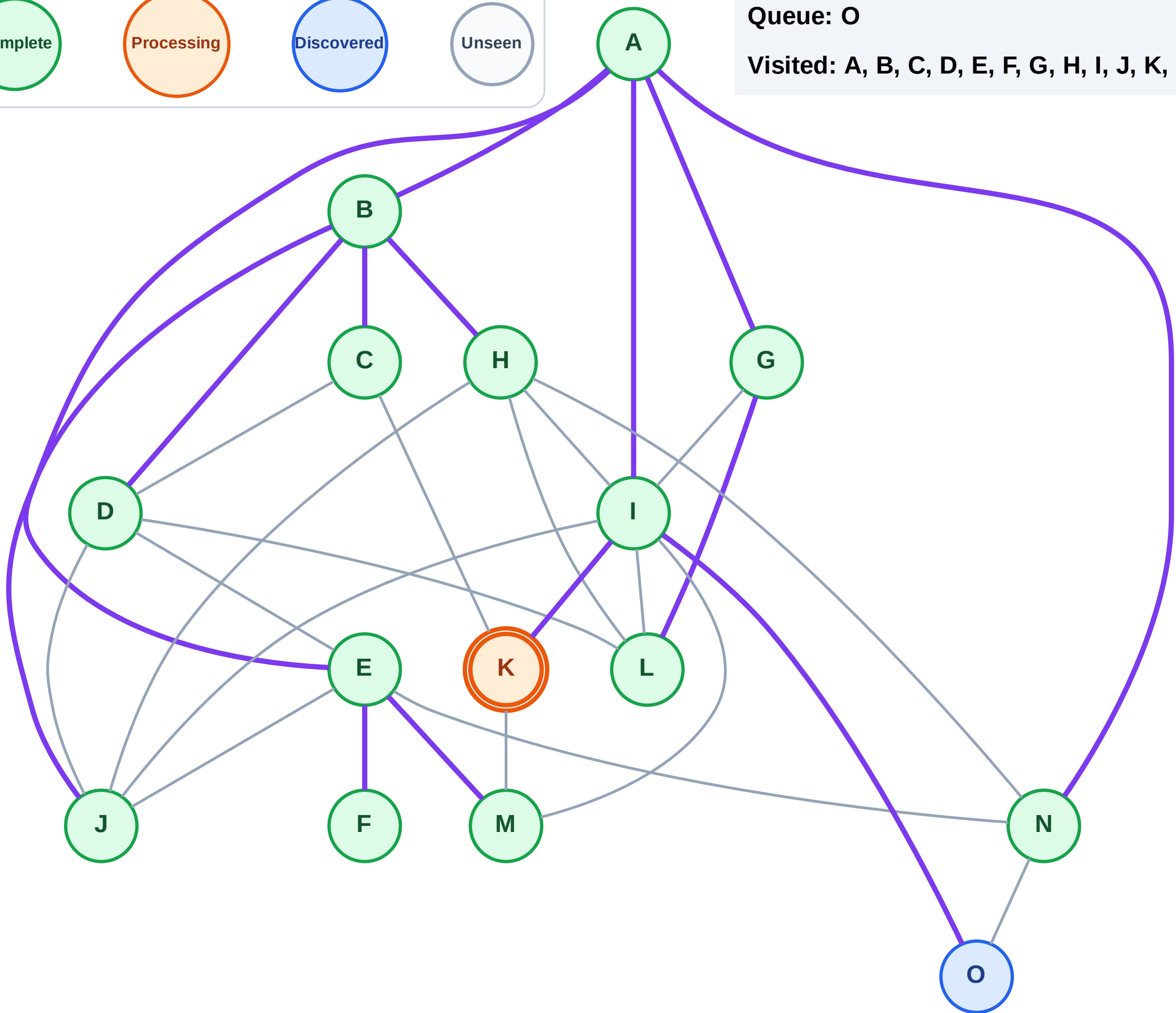
Processing

Discovered

Unseen

Queue: O

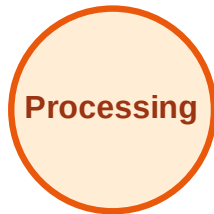
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N



DFS Traversal

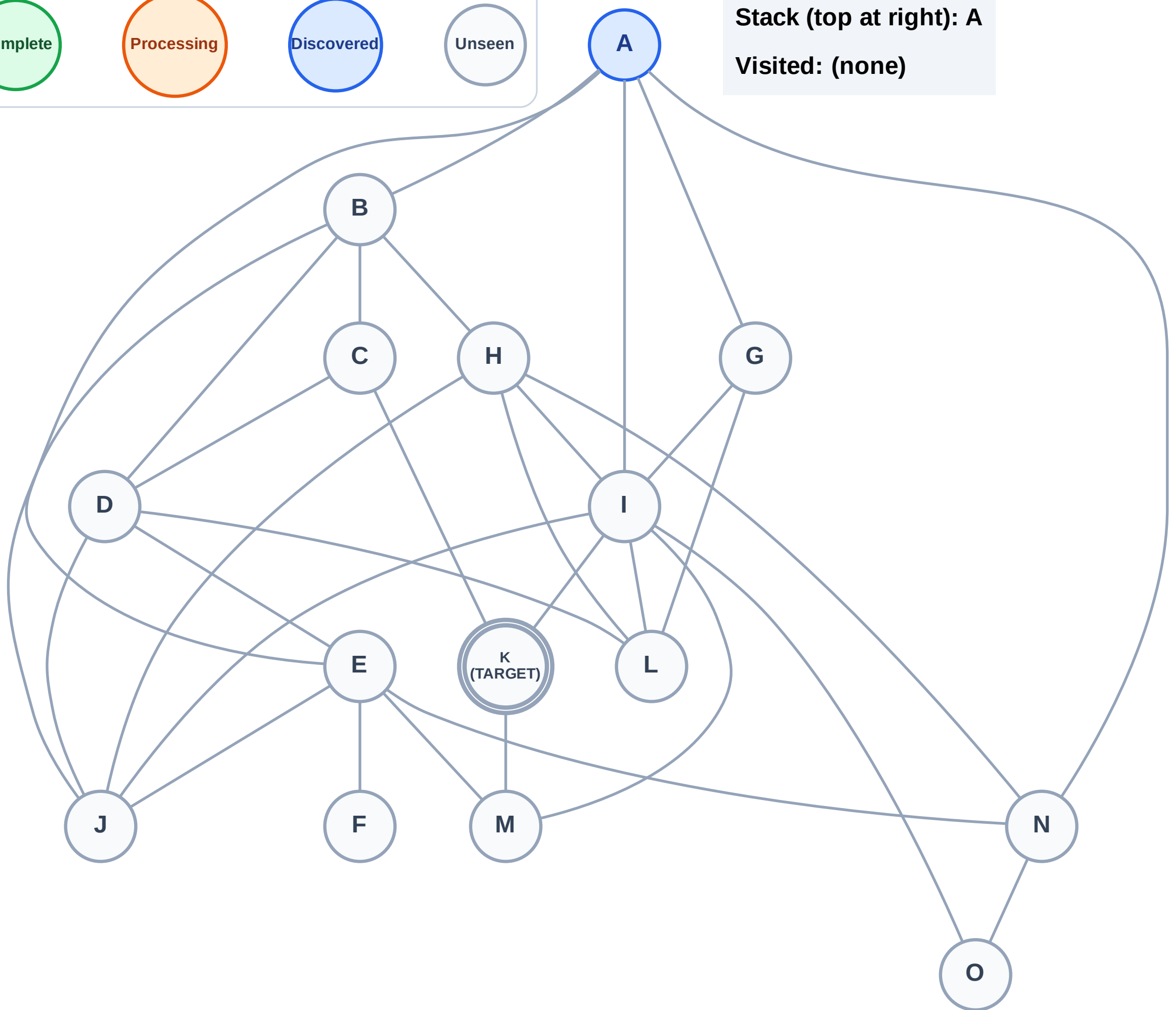
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

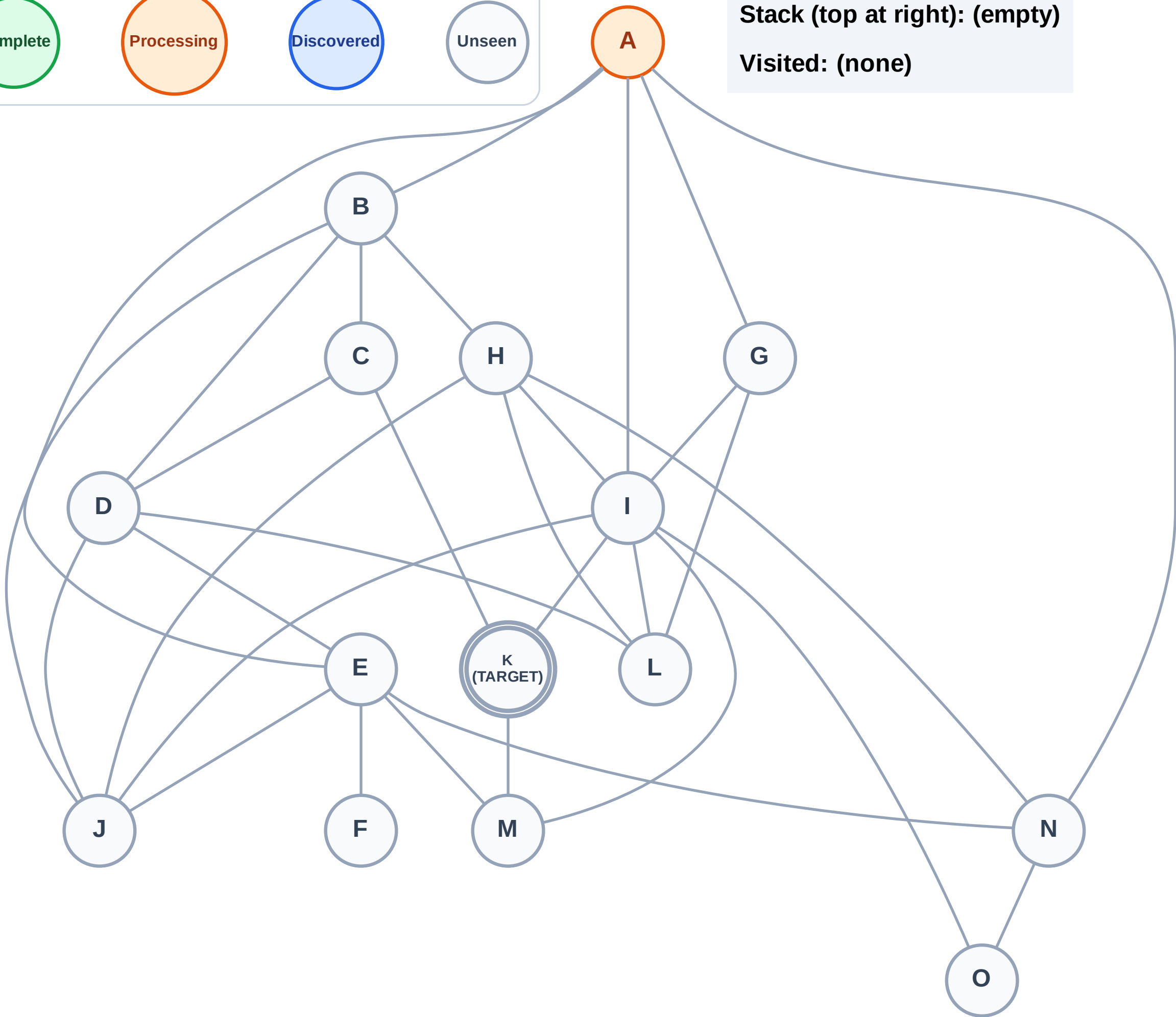
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover N, I, G, E, B from A

Legend

Complete

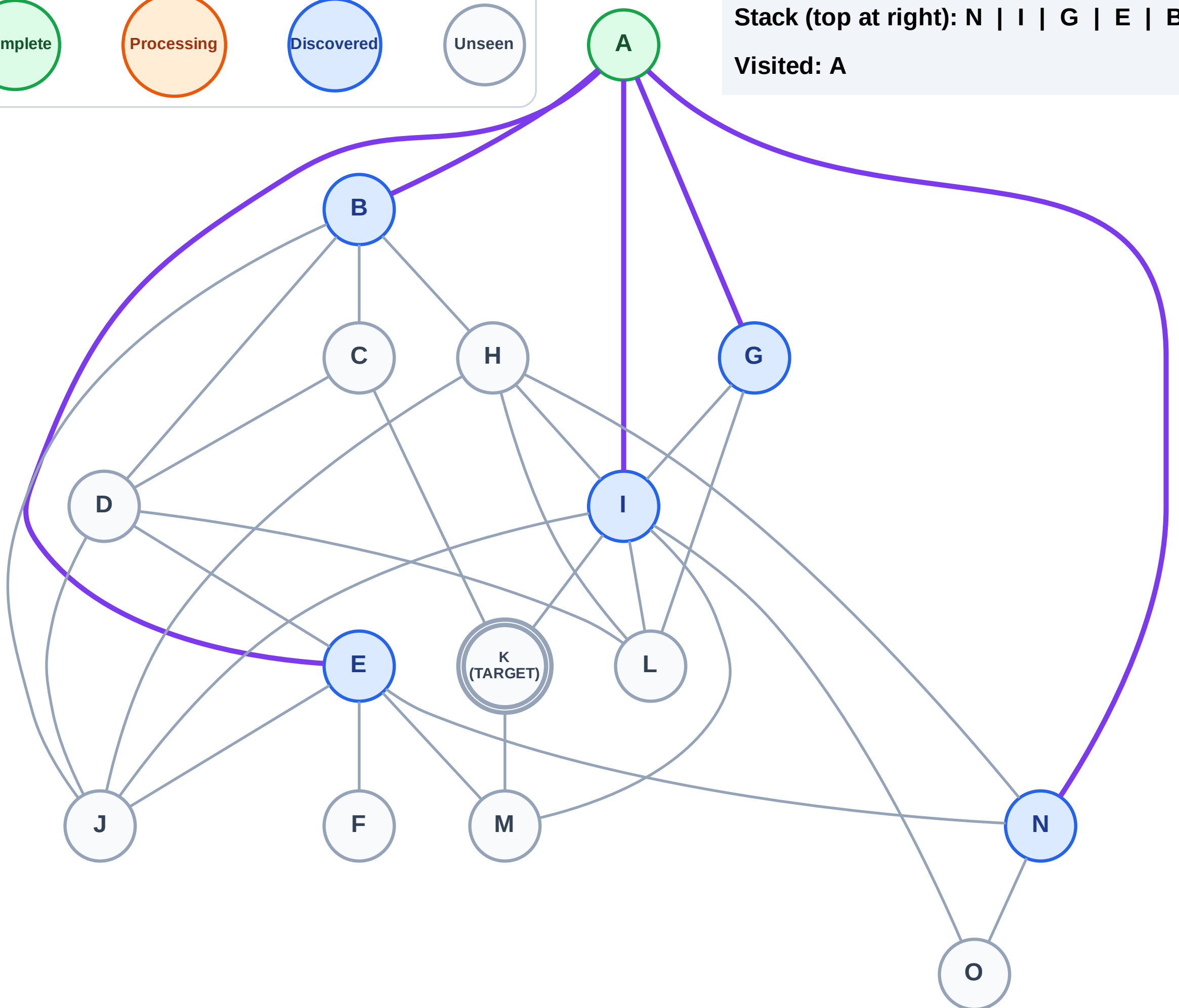
Processing

Discovered

Unseen

Stack (top at right): N | I | G | E | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

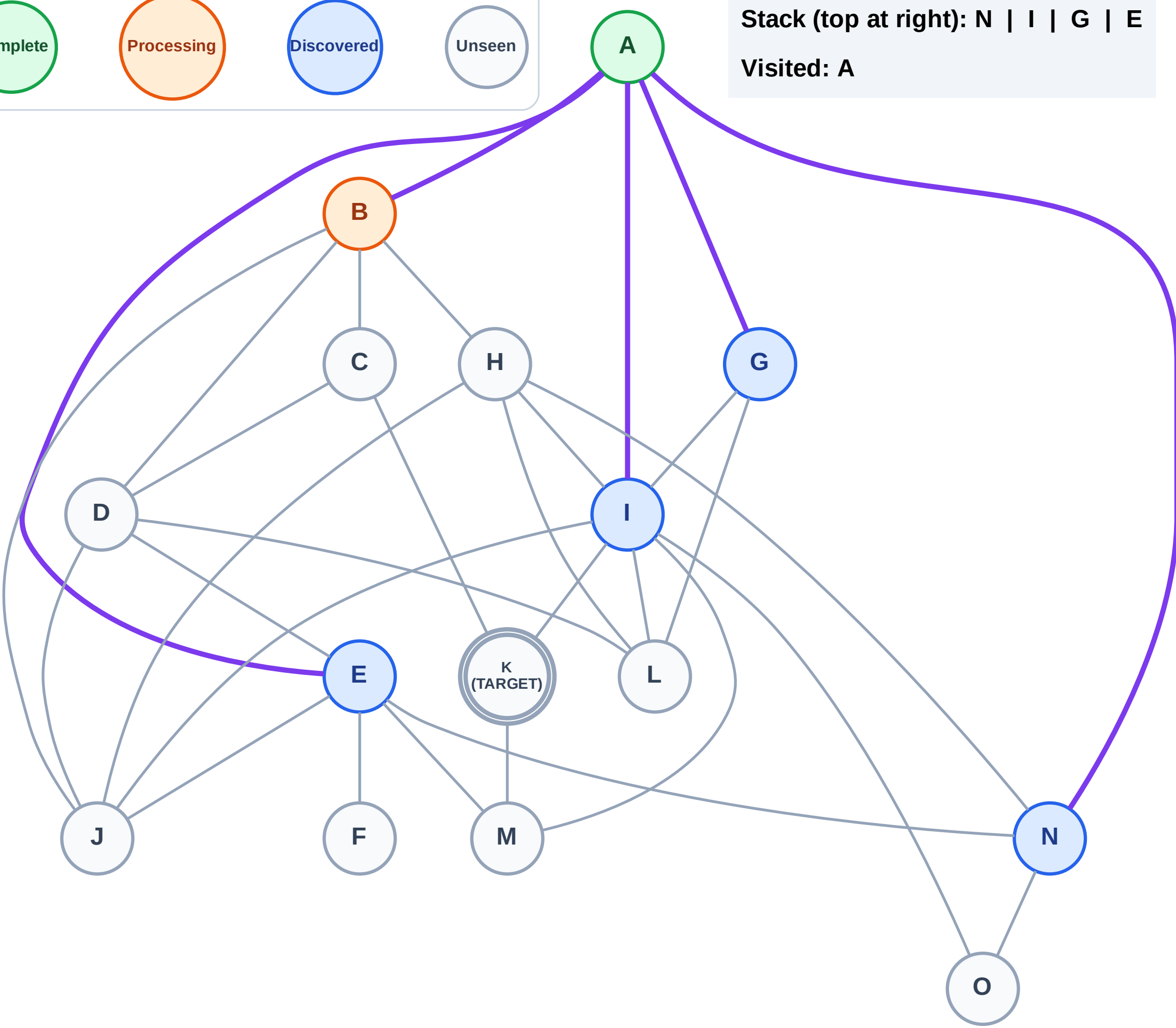
Processing

Discovered

Unseen

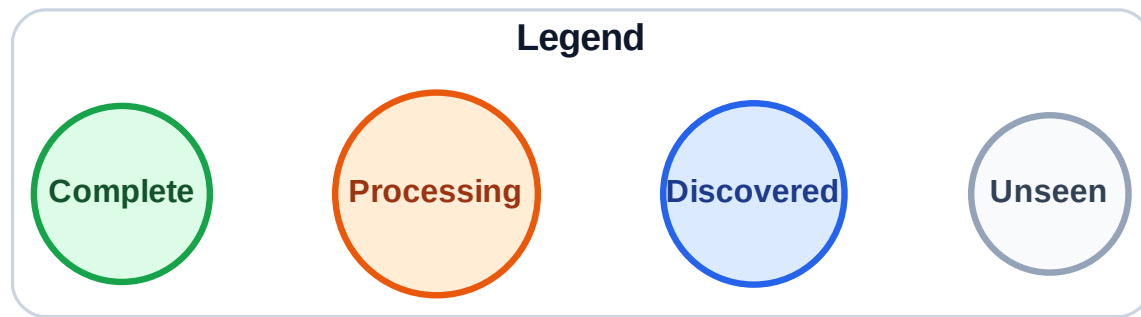
Stack (top at right): N | I | G | E

Visited: A

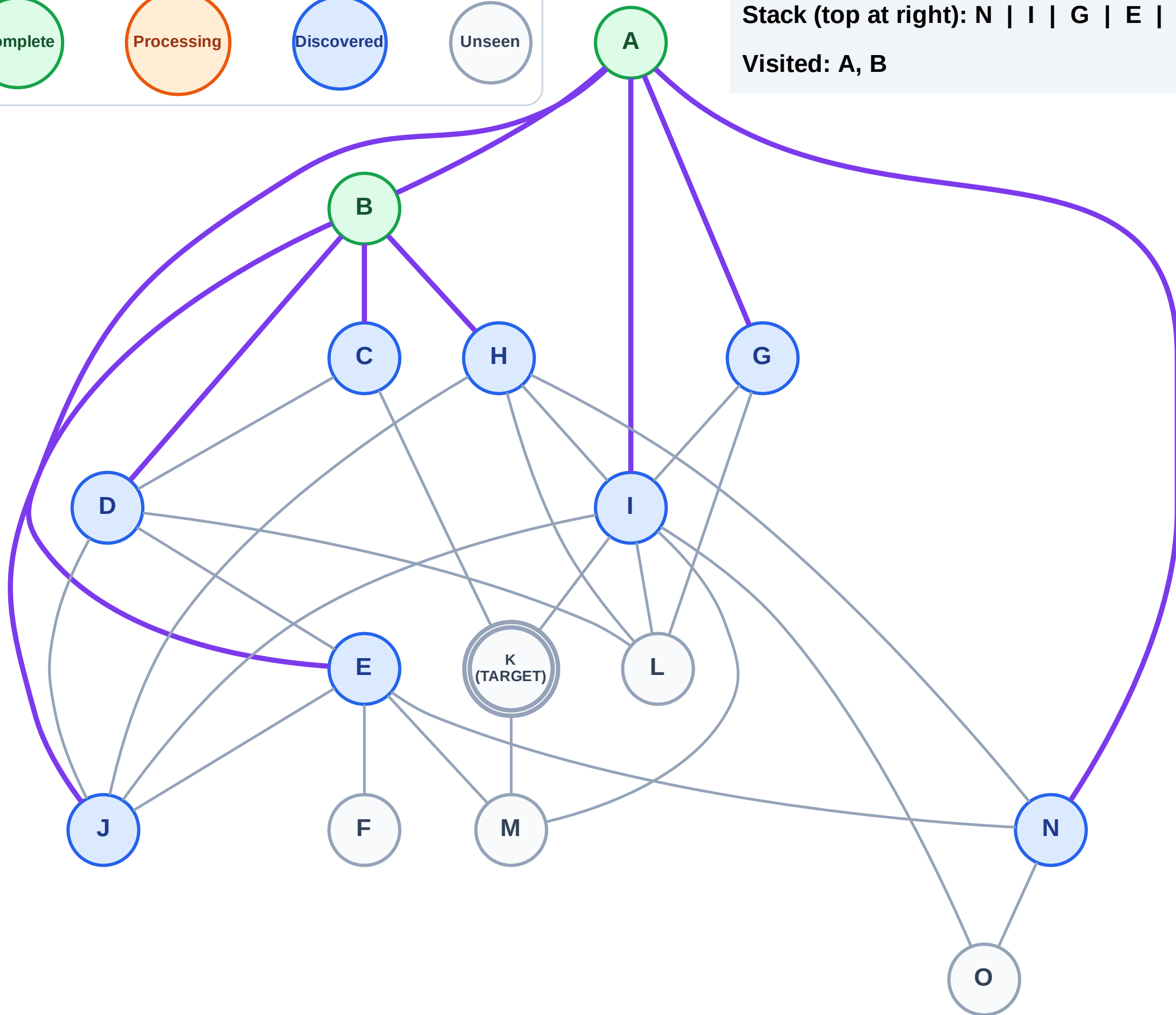


DFS Traversal

Step 5: Discover J, H, D, C from B



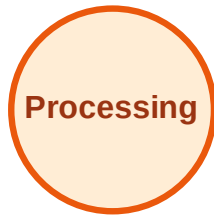
Stack (top at right): N | I | G | E | J | H | D | C
Visited: A, B



DFS Traversal

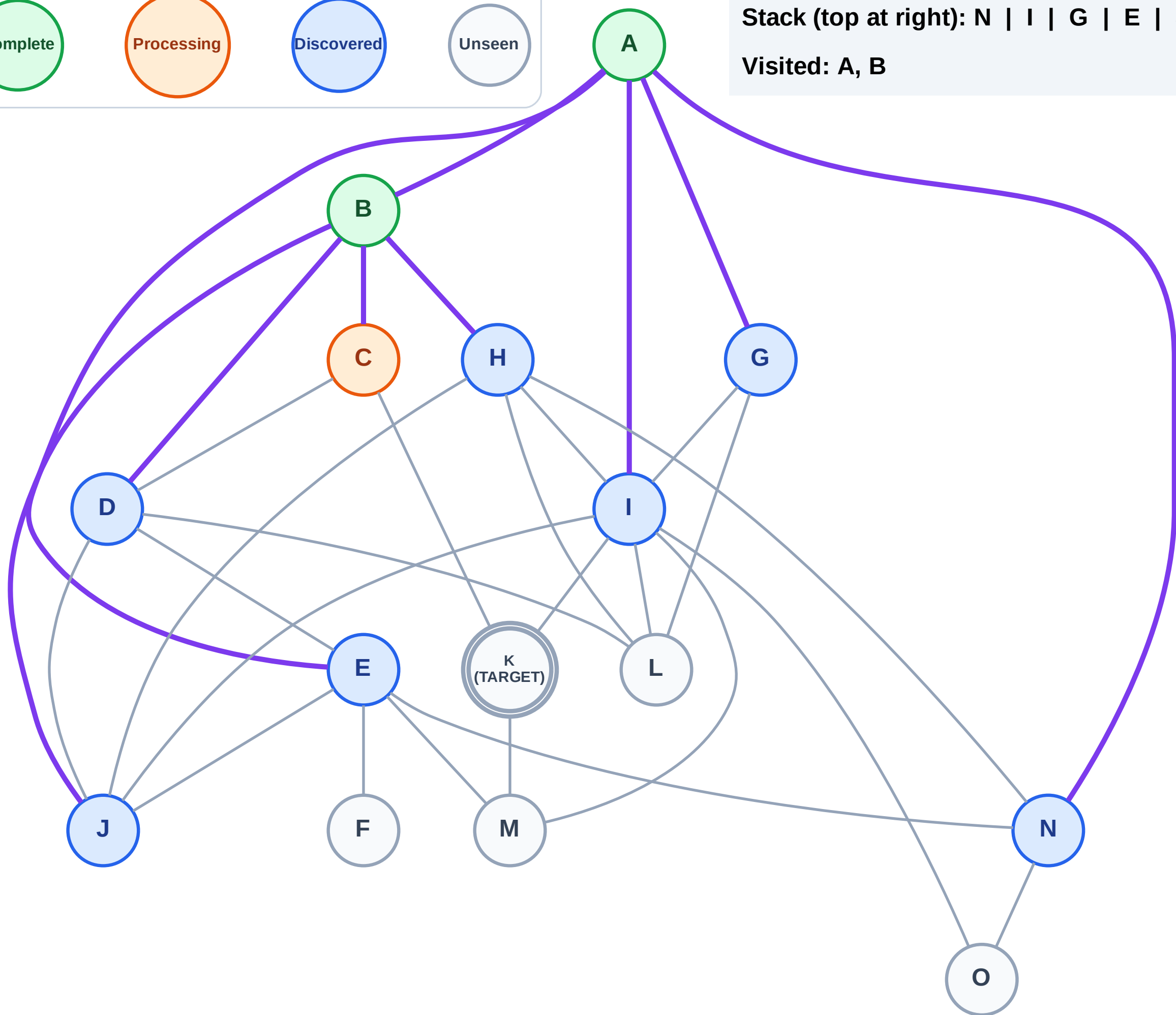
Step 6: Process node C

Legend



Stack (top at right): N | I | G | E | J | H | D

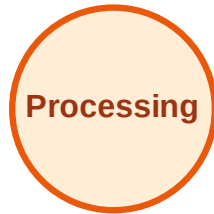
Visited: A, B



DFS Traversal

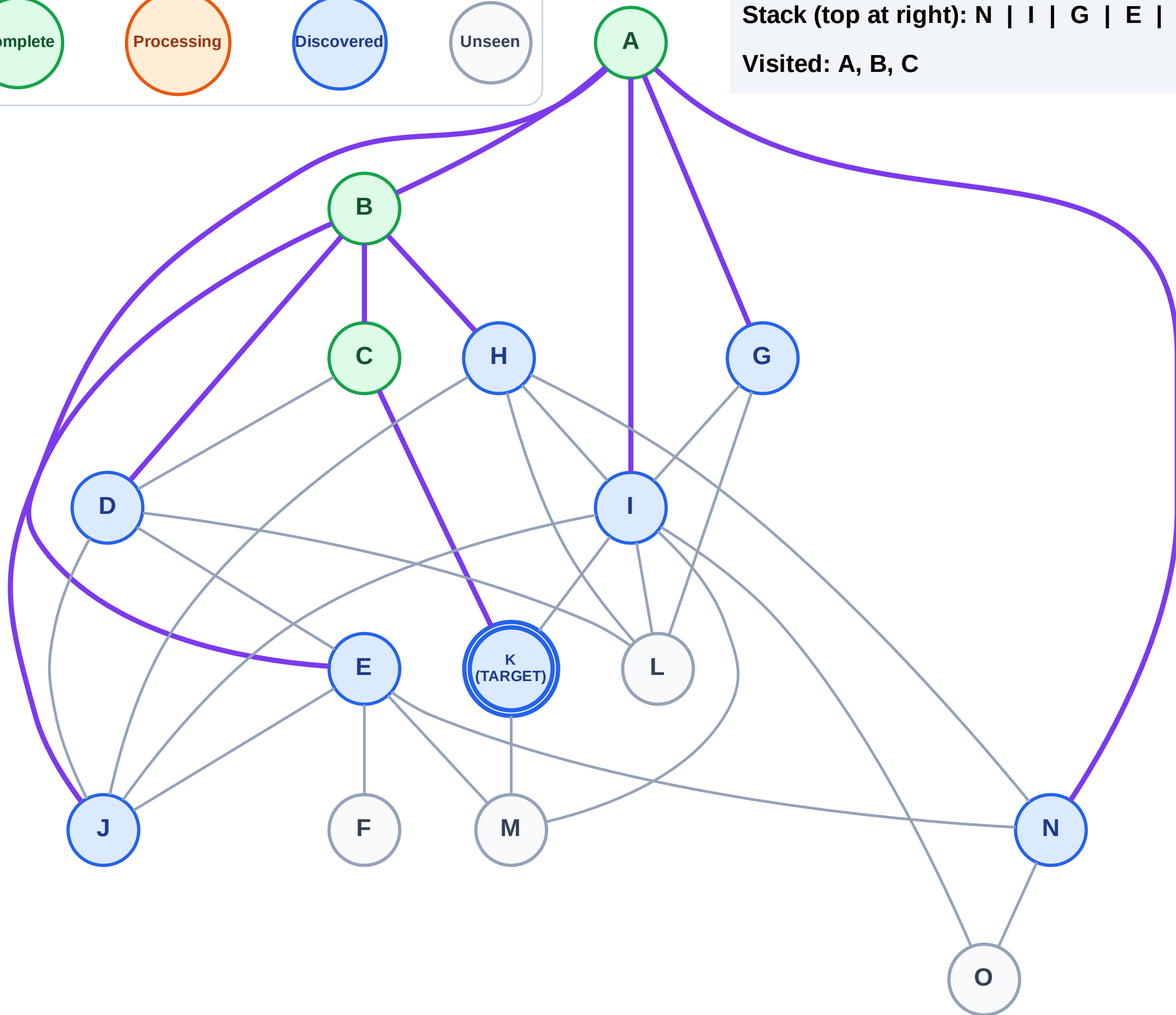
Step 7: Discover K from C

Legend



Stack (top at right): N | I | G | E | J | H | D | K

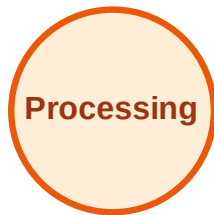
Visited: A, B, C



DFS Traversal

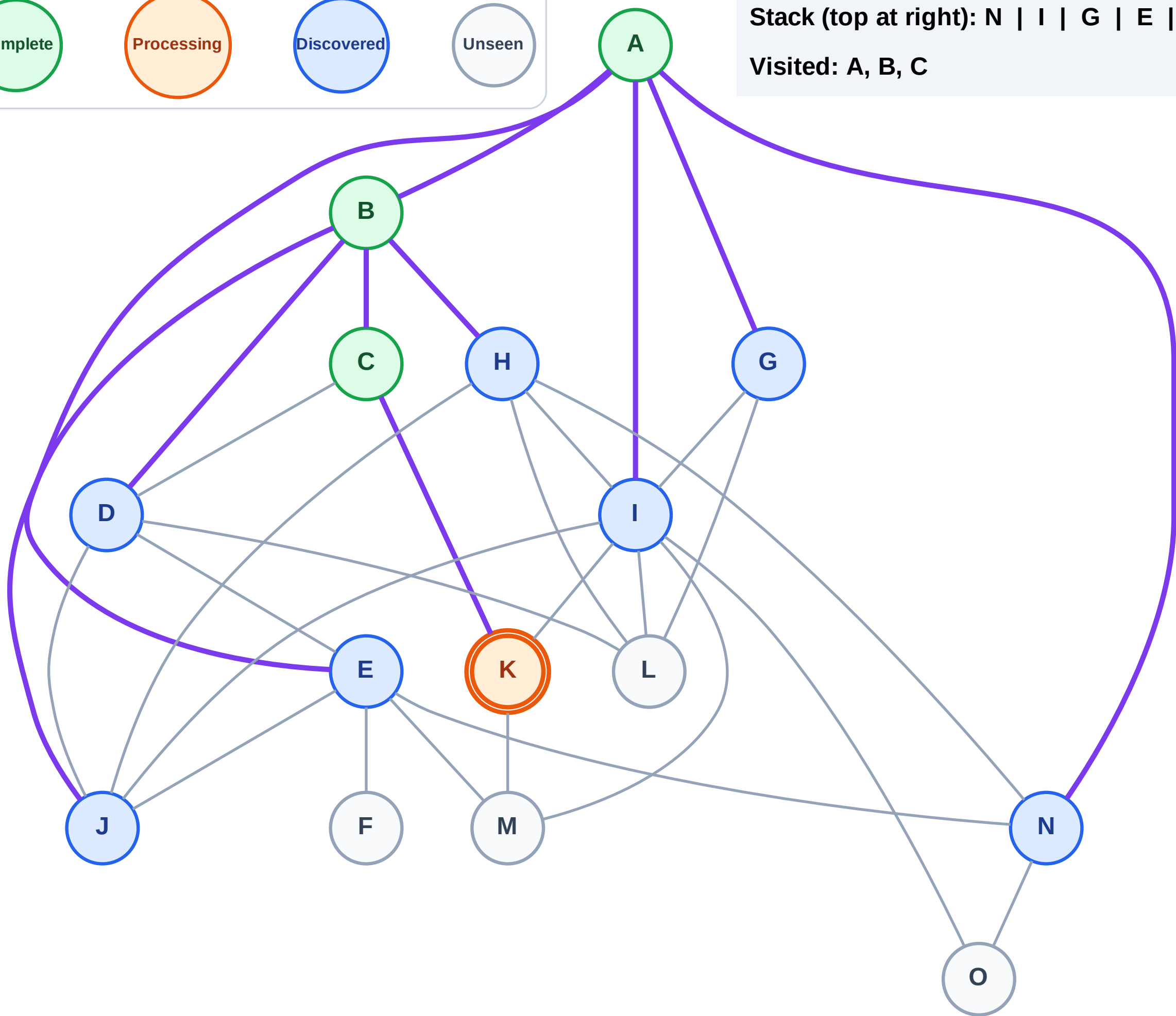
Step 8: Process node K

Legend



Stack (top at right): N | I | G | E | J | H | D

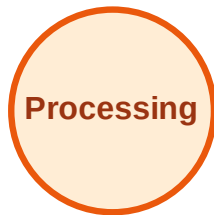
Visited: A, B, C



DFS Traversal

Step 9: TARGET NODE K REACHED! Path (3 hops): A → B → C → K

Legend



Stack (top at right): N | I | G | E | J | H | D

Visited: A, B, C, K

